

Pyruvate and Related Energetic Metabolites Modulate Resilience Against High Genetic Risk for Glaucoma

Reviewed Preprint

v2 • March 28, 2025

Revised by authors

Reviewed Preprint

v1 • March 11, 2025

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eLife Assessment

This study presents a **valuable** finding on the importance of the plasma metabolome in glaucoma risk prediction. The evidence supporting the claims of the authors is **solid** and the work offers insights for the design of protective therapeutic strategies for glaucoma. The authors have addressed the concerns of the reviewers and reported on the limitations of the study.

<https://doi.org/10.7554/eLife.105576.2.sa3>

Abstract

A glaucoma polygenic risk score (PRS) can effectively identify disease risk, but some individuals with high PRS do not develop glaucoma. Factors contributing to this resilience

remain unclear. Using 4,658 glaucoma cases and 113,040 controls in a cross-sectional study of the UK Biobank, we investigated whether plasma metabolites enhanced glaucoma prediction and if a metabolomic signature of resilience in high-genetic-risk individuals existed. Logistic regression models incorporating 168 NMR-based metabolites into PRS-based glaucoma assessments were developed, with multiple comparison corrections applied. While metabolites weakly predicted glaucoma (Area Under the Curve=0.579), they offered marginal prediction improvement in PRS-only-based models ($P=0.004$). We identified a metabolomic signature associated with resilience in the top glaucoma PRS decile, with elevated glycolysis-related metabolites—lactate ($P=8.8E-12$), pyruvate ($P=1.9E-10$), and citrate ($P=0.02$)—linked to reduced glaucoma prevalence. These metabolites combined significantly modified the PRS-glaucoma relationship ($P_{\text{interaction}}=0.011$). Higher total resilience metabolite levels within the highest PRS quartile corresponded to lower glaucoma prevalence (Odds Ratio_{highest vs. lowest total resilience metabolite quartile}=0.71, 95% Confidence Interval=0.64–0.80). As pyruvate is a foundational metabolite linking glycolysis to tricarboxylic acid cycle metabolism and ATP generation, we pursued experimental validation for this putative resilience biomarker in a human-relevant *Mus musculus* glaucoma model. Dietary pyruvate mitigated elevated intraocular pressure ($P=0.002$) and optic nerve damage ($P<0.0003$) in *Lmx1b*^{V265D} mice. These findings highlight the protective role of pyruvate-related metabolism against glaucoma and suggest potential avenues for therapeutic intervention.

Introduction

Glaucoma is a polygenic, progressive neurodegenerative disease and a leading cause of irreversible blindness.¹ The disease is typically asymptomatic until advanced visual field loss occurs, and around 50 to 70% of people affected remain undiagnosed.^{2,3} Early detection and intervention are essential to stop disease progression and prevent visual impairment in glaucoma-affected individuals, as there is no cure available; however, population-based glaucoma screening is not cost-effective from a public health perspective.^{4–6}

Glaucoma is well-suited for developing and applying a polygenic risk score (PRS) to facilitate disease identification and risk stratification as it is a condition with several endophenotypes, such as elevated intraocular pressure (IOP) and thinning of the retinal nerve fiber layer (RNFL), and it is highly heritable.^{7–9} Glaucoma risk is influenced by both genetic and metabolic factors, with emerging evidence suggesting that gene-environment interactions may play a greater role in conferring disease risk than independent exposures alone.^{10–15}

Glaucoma was five times more likely in the UK Biobank (UKBB) among participants with the highest genetic risk decile versus the lowest decile (7.4% vs 1.3%).⁹ While the relatively low glaucoma prevalence in the highest decile group could be explained by disease under-ascertainment or a PRS that incompletely reflects the glaucoma genetic architecture, it is possible that resilience biomarkers could explain this result, in addition to protective genetic or epigenetic factors.

Recent advancements in metabolomics have opened up avenues to explore metabolites as potential biomarkers for glaucoma.¹⁶ Metabolites are intermediate and end products of cellular processes critical for driving cellular growth and tissue homeostasis.¹⁷ These small molecules provide a holistic measure of physiological status, reflecting both genetic predispositions and environmental influences. Previous studies have indicated a potential role for plasma metabolites to stratify glaucoma risk; however, these studies were limited by small sample sizes,^{18–22} with restricted coverage of metabolites. In addition, the potential benefits of the integration of metabolomics with genetics to identify individuals at the highest risk of glaucoma remain unexplored.

This study aims to evaluate plasma metabolites as risk factors for glaucoma. First, we explored whether incorporating plasma metabolite data improved the predictive accuracy of a PRS for glaucoma risk based on available data in the UKBB, where genetic and metabolomic data were available for 117,698 participants. Additionally, we evaluated interactions between glaucoma PRS and metabolomics for further stratifying individuals with high genetic risk but without glaucoma. We undertook an agnostic approach to identify a metabolite signature associated with resilience to high glaucoma genetic risk. Finally, we experimentally validated a top resilience metabolite by assessing its ability to rescue the ocular phenotype in a human-relevant, genetic mouse model of glaucoma.

Results

UK Biobank Study Characteristics

A total of 117,698 participants (4,658 glaucoma cases and 113,040 non-cases) were included in this study. UKBB participants were predominately of European ancestry (85.8%) but were also of admixed American (0.22%), Asian (2.77%), and African (1.73%) ancestry. The design of the human studies is depicted graphically in **Figures 1** and **2**. There were significant differences seen in baseline demographic and clinical characteristics between the glaucoma cases and the non-cases presented in **Table 1**. Notably, characteristics associated with glaucoma included male gender, older age, genetic African ancestry (Privé et al.²³ describes the method for determining genetically-inferred ancestry in the UKBB), prior history of smoking, slightly lower cholesterol levels, higher body mass index (BMI), higher hemoglobin A1c (HbA1c), higher IOP, thinner macular retinal nerve fiber layer (mRNFL) thickness, higher caffeine intake, higher alcohol intake, higher oral steroid use, diabetes, and coronary artery disease.

Plasma metabolites marginally improve glaucoma risk prediction

We first sought to determine whether the addition of metabolites could improve the prediction of glaucoma from basic demographics, clinical variables, and genetic data. We created four logistic regression models, each with increasing predictive variables considered, to evaluate their added utility for predicting glaucoma (see Methods and Supplemental Table 2 for model construction details). We analyzed two sets of metabolite variables: a comprehensive set of 168 metabolites measured by nuclear magnetic resonance (NMR) spectroscopy included in the UKBB dataset and a limited set of 27 metabolites approved by the European Union (EU) for *in vitro* diagnostic use.

To evaluate the performance of each model, we plotted receiver operating characteristic (ROC) curves and calculated the area under the curve (AUC) metrics. In model 1, which utilized only metabolite data, the highest performance was achieved using the full panel of 168 metabolites, yielding an AUC value of 0.602 (95% CI=0.592-0.612) compared to 0.579 (95% CI=0.569-0.589; $P=0.0003$) using the 27 metabolites (**Figure 3**). Model 2, which incorporated demographic information, and model 3, which also included clinical variables, exhibited similar performance before the addition of metabolite data. In model 2, the addition of the 168 metabolite panel demonstrated the best performance, with an AUC value of 0.670 (95% CI=0.660-0.680) compared to 0.664 (95% CI=0.654-0.674) for the model without metabolites ($P=0.002$) and 0.666 (95% CI=0.656-0.676) for the limited panel of 27 metabolites. This performance trend persisted in model 3 with the inclusion of 168 metabolites, producing an AUC of 0.680 (95% CI=0.660-0.700), representing an increase in AUC ($P=0.02$) from model 3 with the exclusion of metabolites (AUC 0.670; 95% CI=0.650-0.690). We subsequently examined whether integrating metabolite data into a PRS-based model could improve glaucoma prediction algorithms. In model 4, the panel of 168 metabolites yielded the best performance with an AUC value of 0.806 (95% CI=0.796-0.816), while the model without metabolites (AUC=0.801, 95% CI=0.791-0.811; $P=0.004$), as well as the model with a limited panel of

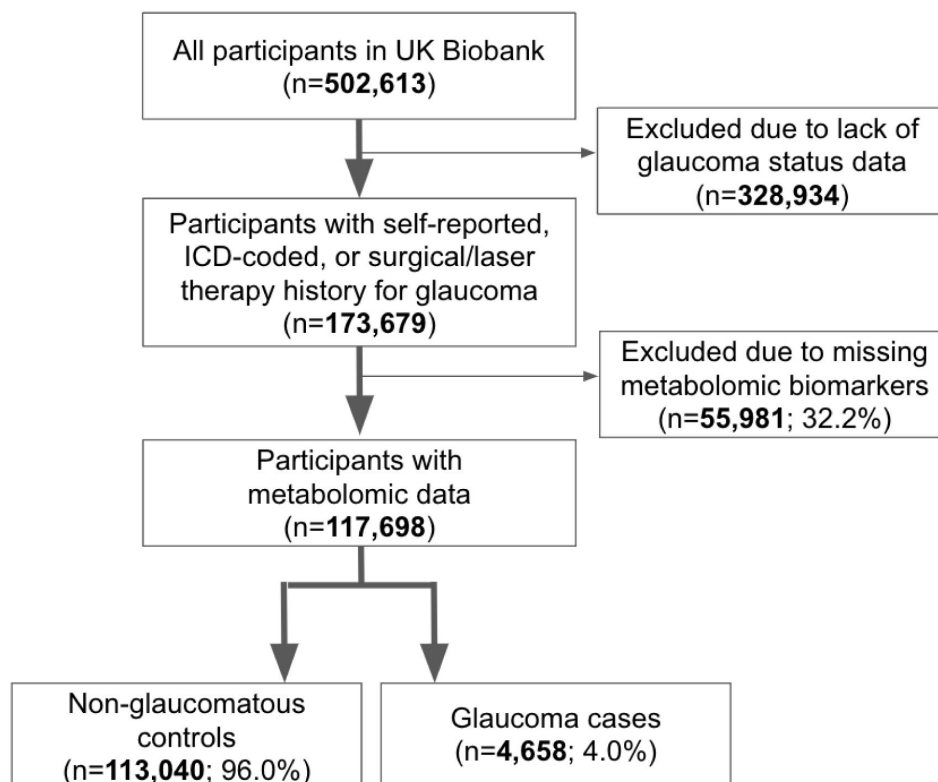


Figure 1.

Participant flow chart describing inclusion and exclusion criteria from the UK Biobank

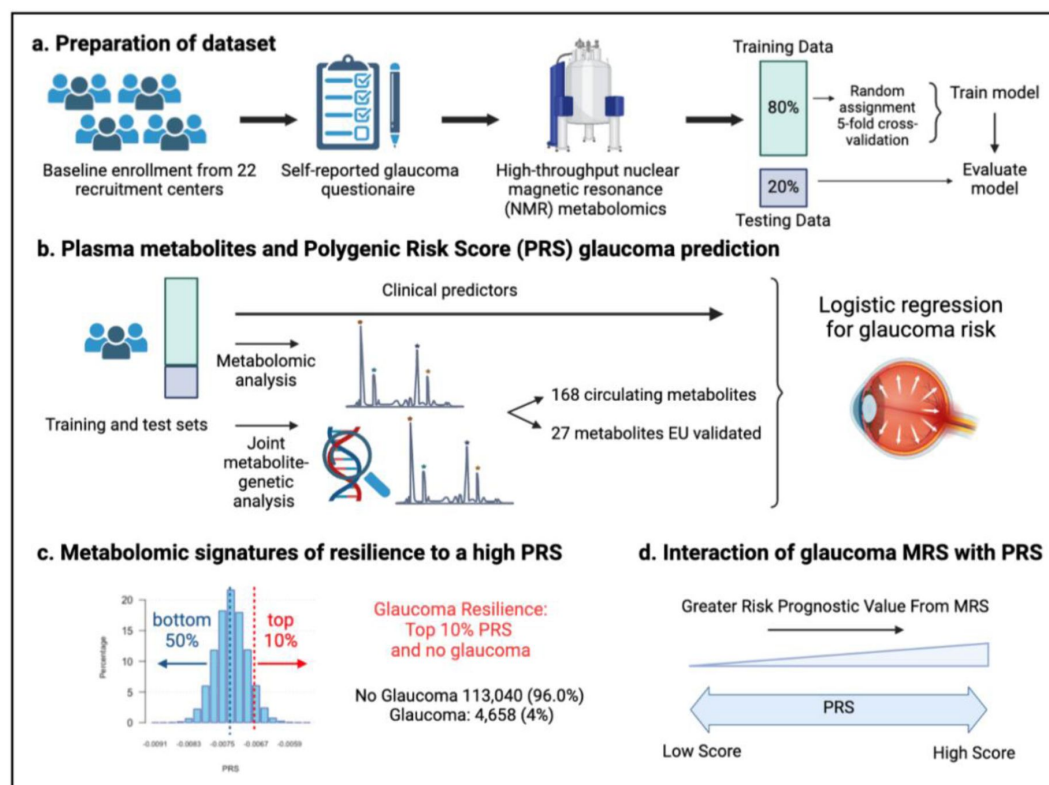


Figure 2.

Study design from the UK Biobank.

(a) 117,698 individuals had metabolomics data available from the UK Biobank, which was divided into a training and test set to formulate a metabolic risk score (MRS) model. (b) The inclusion of metabolites (either 168 metabolites on the NMR platform or a subset of 27 metabolites with European Union (EU) certification) in relation to prevalent glaucoma risk prediction was studied. (c) A histogram showing the polygenic risk score (PRS) distribution is shown. Overall, 4,658 cases and 113,040 individuals without glaucoma are available for analysis. The metabolomic signature of resilience to the top 10% of glaucoma PRS was assessed among 1,693 cases (14.4%) and 10,077 individuals without glaucoma (85.6%). (d) Interactions of prevalent glaucoma with MRS and PRS quartiles were examined.

Characteristic	No Glaucoma	Glaucoma	P-value
Sample Size (%)	113,040 (96.0)	4,658 (4.0)	
Sex - Male (%)	52,497 (46.4)	2,493 (53.5)	<0.001
Age at recruitment (mean (SD))	56.7 (8.0)	60.9 (6.6)	<0.001
Ethnicity (%)			0.0010
White	105,912 (94)	4,331 (93)	
Asian	2,977 (2.6)	127 (2.7)	
Black	2,228 (2.0)	130 (2.8)	
Other	1,923 (1.7)	70 (1.5)	
Genetic Ancestry (%)			<0.001
African	1,911 (1.7)	123 (2.7)	
AMR	261 (0.2)	2 (0.0)	
Asian	3,125 (2.8)	134 (2.9)	
European	96,991 (86.6)	4,001 (86.7)	
Smoking Status (%)			<0.001
Never	63,240 (56)	2,339 (52)	
Prefer not to answer	430 (0.4)	16 (0.4)	
Previous	38,926 (34)	1,732 (39)	
Current	10,605 (9.4)	410 (9.1)	
Total Cholesterol, mmol/l (median [IQR])	4.6 [3.98, 5.22]	4.5 [3.88, 5.18]	<0.001
Physical Activity, MET-minutes per week (mean (SD))	2,465 (2430)	2,458 (2,438)	0.84
Body Mass Index, kg/m ² (mean (SD))	27.4 (4.8)	27.7 (4.7)	<0.001
HbA1c, mmol/mol (mean (SD))	36.0 (5.8)	37.4 (6.8)	<0.001
Spherical Equivalent, diopter (mean (SD))	-0.1 (2.1)	-0.1 (1.9)	0.67
Intraocular pressure, mmHg (mean (SD))	15.9 (2.6)	17.8 (4.5)	<0.001
mRNFL thickness, μ m (mean (SD))	28.7 (1.8)	28.5 (1.3)	<0.001
Beta blocker use (%)	8,398 (7.4)	447 (9.6)	<0.001
Caffeine intake, mg/day (mean (SD))	165 (67)	168 (59)	0.0020
Alcohol intake, g/week (median [IQR])	84 [40.3, 146.9]	84 [48.0, 154.6]	<0.001
Diabetes (%)	6,512 (5.8)	463 (9.9)	<0.001
Oral steroid use (%)	3,043 (2.7)	184 (4.0)	<0.001
Coronary Artery Disease (%)	5,153 (4.6)	377 (8.1)	<0.001

Abbreviations: SD, standard deviation; IQR, interquartile range; AMR, mixed American; MET metabolic equivalent of task mRNFL, macular retinal nerve fiber layer

Table 1.

Demographic and clinical characteristics of the UK Biobank study population assessed in 2006-2010

27 metabolites (AUC=0.802, 95% CI=0.792-0.812; $P=0.006$) both had lower AUC values. Thus, these findings suggest that the addition of metabolite data marginally enhances the prediction of glaucoma beyond the use of demographic and genetic data.

We then investigated whether the inclusion of the 168 metabolites could improve the predictive value for glaucoma by stratifying patients into specific subgroups and calculating AUC curves for each stratum. Accordingly, we found that metabolites showed a modestly improved predictive value for glaucoma among people of White ethnicity ($P<0.001$), those aged 55 and older ($P=0.002$), and males ($P=0.002$) (**Table 2** [↗](#)).

Metabolites associated with resilience to a high glaucoma polygenic risk score

As metabolite data only marginally augmented clinical or PRS predictions of glaucoma, we hypothesized that metabolites might provide utility for differentiating patients within risk groups. Specifically, we studied participants who possessed a high glaucoma PRS but did not have glaucoma. For this study, we labeled them resilient while recognizing there are many reasons they may not have glaucoma. Participants were stratified into PRS deciles, and metabolite signatures were identified to differentiate patients with and without glaucoma in the top decile ($N=11,770$) and the bottom half ($N=58,358$) of the glaucoma PRS distribution (**Figure 4** [↗](#)). Within the top decile of glaucoma PRS, compared to participants without glaucoma, participants with glaucoma were more likely to be older, male, of White ethnicity, and were prior smokers. Glaucoma participants with the highest PRS also had higher BMI, higher HbA1c, higher spherical equivalent, consumed more caffeine and alcohol, and were more likely to have diabetes and coronary artery disease (**Table 3** [↗](#)). Among participants in the bottom half of glaucoma PRS, participants with glaucoma were also more likely to be older, male, of Black and Asian ethnicity, prior smokers, had higher BMI and higher HbA1c, and were more likely to have diabetes and coronary artery disease (**Table 4** [↗](#)). As expected, participants with glaucoma in both bins of glaucoma genetic risk had higher IOP and thinner mRNFL thickness.

Our multivariable-adjusted analysis revealed that higher probit-transformed levels of lactate (adjusted $P_{NEF}=8.8E-12$), pyruvate (adjusted $P_{NEF}=2.9E-10$), and citrate (adjusted $P_{NEF}=0.018$) were independently associated with no glaucoma in the top decile of glaucoma PRS (**Table 5** [↗](#)). In addition, lower levels of triglycerides and higher levels of selected HDL analytes had an adjusted $P_{NEF} < 0.2$ and were associated with no glaucoma among participants with high genetic risk. Among the bottom half of the PRS distribution, higher levels of albumin were associated with no glaucoma (adjusted $P_{NEF}=0.047$), while higher levels of small HDL, omega-3 fatty acids, docosahexaenoic acid, lactate, and citrate were associated with no glaucoma, albeit with an adjusted $P_{NEF} < 0.2$.

Interaction of metabolic and genetic biomarkers

The three metabolites (lactate, pyruvate, and citrate) for which higher levels were associated with reduced glaucoma prevalence in individuals with high genetic susceptibility had no statistically significant relationship with glaucoma for participants in the lower half of the PRS distribution at the adjusted $P_{NEF} < 0.05$ level. Thus, we hypothesized that there may be an interaction between these metabolites and genetic risk. Such an interaction would suggest that these glycolysis/tricarboxylic acid (TCA) cycle metabolites are relevant primarily in the setting of high PRS, establishing them as resilience factors against elevated genetic risk of glaucoma. Indeed, we observed a significant interaction between elevated levels of total lactate, pyruvate, and citrate with glaucoma PRS quartile for predicting the risk of glaucoma ($P_{interaction}=0.011$; **Figure 5A** [↗](#)).

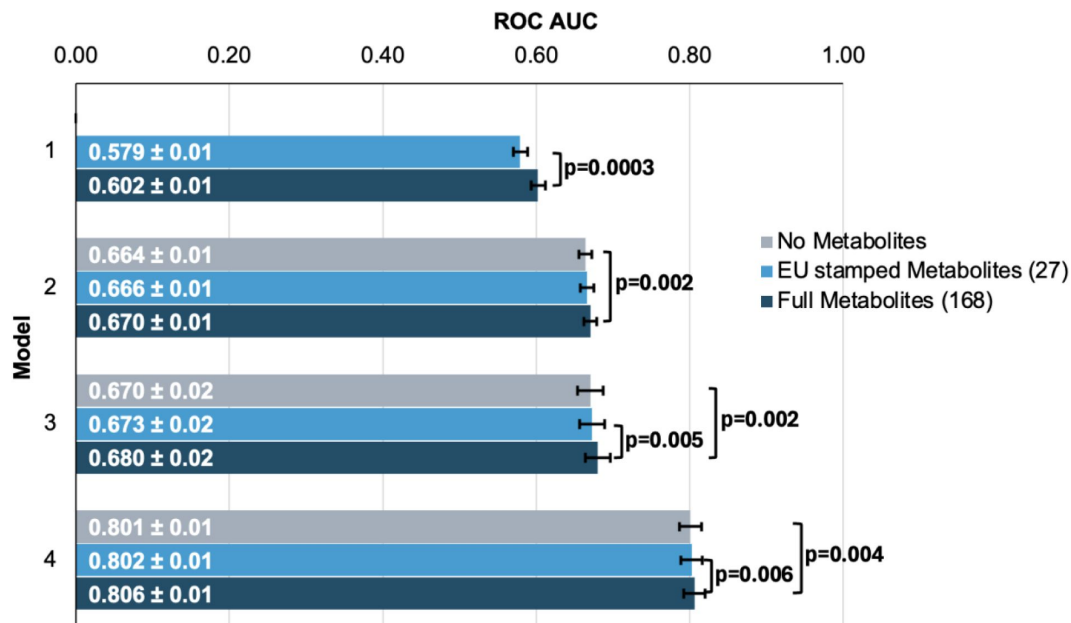


Figure 3.

Inclusion of metabolite data into glaucoma prediction algorithms.

Model 1 includes metabolites only; Model 2 incorporates additional covariates including age, sex, genetic ancestry, season, time of day of specimen collection, and fasting time; Model 3 incorporates covariates in Model 2 and smoking status (never, past, and current smoker), alcohol intake (g/week), caffeine intake (mg/day), physical activity (metabolic equivalent of task [MET], hours/week), body mass index (kg/m²), average systolic blood pressure (mm Hg), history of diabetes, HbA1c (mmol/mol), history of coronary artery disease, systemic beta-blocker use, oral steroid use, and spherical equivalent refractive error (diopters); Model 4 incorporates covariates in model 3 and a glaucoma polygenic risk score (PRS). Each color represents a different panel of metabolites (grey = no metabolites; light blue = 27 metabolites; and dark blue = 168 metabolites). The white text represents the AUC ± 95% confidence interval.

Abbreviations: ROC, receiver operator curve; AUC, area under the curve; EU, European Union.

Stratification	Sample size	Glaucoma cases	No Metabolites (AUC)	Metabolites (AUC)	P-value
Ethnicity					
White	110,243	4,331	0.675	0.686	<0.001
Asian	3,104	127	0.780	0.768	0.062
Black	2,358	130	0.704	0.706	0.52
Age					
<55 years	43,648	788	0.576	0.566	0.52
≥55 years	74,050	3,870	0.569	0.596	0.002
Gender					
Female	62,708	2,165	0.689	0.696	0.25
Male	54,990	2,493	0.659	0.673	0.002

The area under the curve (AUC) of the receiver operating characteristic (ROC) curve was calculated for each demographic stratification to evaluate the predictive performance of models both with and without metabolite data. Differences in model AUC were tested using the DeLong test, and p-values were reported. For models excluding metabolite data, the predictors include as appropriate, age, sex, genetic ancestry, season, time of day of specimen collection, fasting time (hours), smoking status (never, past, and current smoker), alcohol intake (g/week), caffeine intake (mg/day), physical activity (metabolic equivalent of task [MET], hours/week), body mass index (kg/m²), average systolic blood pressure (mm Hg), history of diabetes, HbA1c (mmol/mol), history of coronary artery disease, systemic beta-blocker use, oral steroid use, and spherical equivalent refractive error (diopters). Models including metabolite data incorporated the same predictors with the addition of the 168 metabolite measurements.

Table 2.

Stratification of glaucoma by ethnicity, age, and gender for predictive assessment with and without using metabolite data.

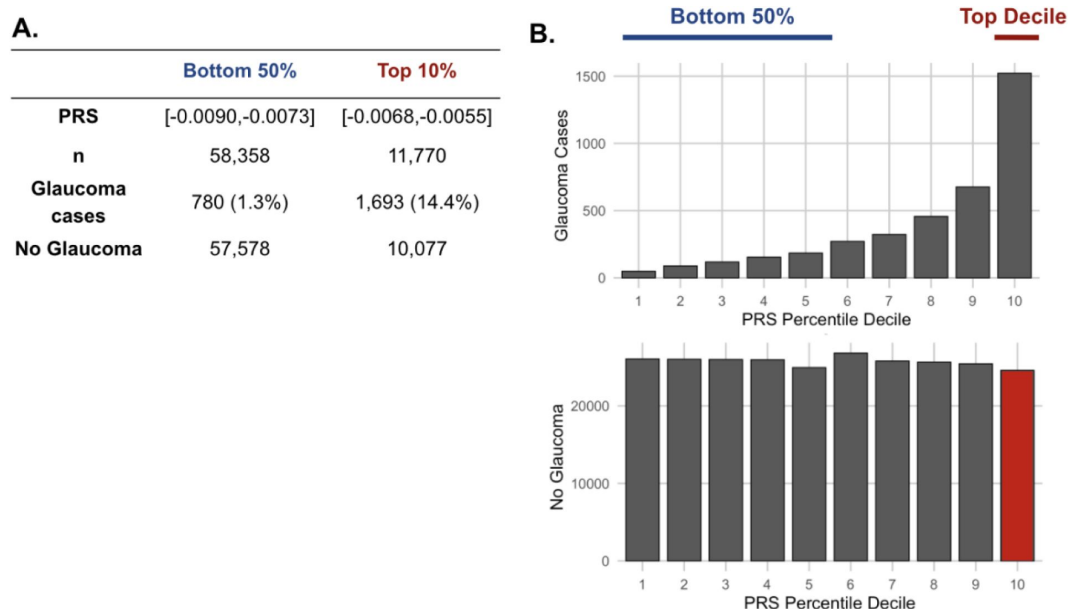


Figure 4.

The distribution of glaucoma cases and no glaucoma stratified by polygenic risk score (PRS) deciles.

Participants were divided into the bottom 50% and top 10% based on their glaucoma PRS, where the prevalence of glaucoma cases from (A) the bottom 50% (n=58,358) was 1.3% and from the top 10% (n=11,770) was 14.4%. (B) Box plot illustrating the distribution of participants with glaucoma (top) and no glaucoma (bottom) as a function of PRS decile. The blue line denotes participants at the bottom 50% of glaucoma PRS, the red line highlights the participants at the top decile of glaucoma PRS, and the red box represents the participants resilient to glaucoma despite high PRS.

Table 3.

Demographic and clinical characteristics in 2006-2010 of participants among the top 10% of glaucoma polygenic risk score (PRS).

Characteristic	No Glaucoma	Glaucoma	P-value
Sample size (%)	10,077 (85.6)	1,693 (14.4)	
Sex - Male (%)	4,667 (46.3)	893 (52.7)	<0.001
Age at recruitment (mean (SD))	56.3 (8.0)	61.0 (6.3)	<0.001
Ethnicity (%)			<0.001
White	9,256 (91.9)	1,636 (96.6)	
Asian	279 (2.8)	24 (1.4)	
Black	345 (3.4)	20 (1.2)	
Other	197 (2.0)	13 (0.8)	
Smoking Status (%)			0.0030
Never	5,806 (57.6)	899 (53.1)	
Prefer not to answer	48 (0.5)	5 (0.3)	
Previous	3,402 (33.8)	640 (37.8)	
Current	821 (8.1)	149 (8.8)	
Physical Activity, MET-minutes per week (mean (SD))	2,439 (2,352)	2,514 (2,480)	0.23
Body mass index, kg/m ² (mean (SD))	27.4 (4.7)	27.7 (4.8)	0.0080
HbA1c, mmol/mol (mean (SD))	36.00 (5.9)	37.0 (6.1)	<0.001
Spherical Equivalent, diopter (mean (SD))	-0.3 (2.2)	-0.2 (1.9)	0.0090
Intraocular pressure, mmHg (mean (SD))	17.2 (3.0)	18.3 (5.0)	<0.001
mRNFL thickness, μ m (mean (SD))	28.7 (1.8)	28.5 (1.2)	<0.001
Beta blocker use (%)	719 (7.1)	179 (10.6)	<0.001
Caffeine intake, mg/day (mean (SD))	164 (65)	170 (60)	0.0010
Alcohol intake, g/week (median [IQR])	84 [38.6, 145.3]	84 [51.5, 155.3]	0.002
Diabetes mellitus (%)	607 (6.0)	151 (8.9)	<0.001
Oral steroid use (%)	292 (2.9)	59 (3.5)	0.22
Coronary Artery Disease (%)	443 (4.4)	148 (8.7)	<0.001
Total cholesterol, mmol/l (median [IQR])	4.61 [4.00, 5.25]	4.54 [3.90, 5.21]	0.006
Lactate, mmol/l (median [IQR])	3.96 [3.27, 4.78]	3.79 [3.16, 4.58]	<0.001
Pyruvate, mmol/l (median [IQR])	0.080 [0.06, 0.10]	0.077 [0.06, 0.09]	<0.001
Citrate, mmol/l (median [IQR])	0.065 [0.06, 0.07]	0.065 [0.06, 0.07]	0.87
Cholesteryl Esters in Small HDL, mmol/l (median [IQR])	0.33 [0.30, 0.36]	0.33 [0.30, 0.36]	0.005
Triglycerides in Very Large VLDL, mmol/l (median [IQR])	0.091 [0.05, 0.15]	0.10 [0.05, 0.17]	<0.001
Alanine, mmol/l (median [IQR])	0.29 [0.24, 0.35]	0.29 [0.24, 0.35]	0.58
Triglycerides in Chylomicrons and extremely Large VLDL, mmol/l (median [IQR])	0.088 [0.03, 0.18]	0.10 [0.04, 0.21]	<0.001
Acetoacetate, mmol/l (median [IQR])	0.010 [0.01, 0.02]	0.011 [0.01, 0.02]	<0.001
Cholesteryl Esters in Medium HDL, mmol/l (median [IQR])	0.41 [0.34, 0.48]	0.40 [0.34, 0.47]	0.003
Triglycerides in Large VLDL, mmol/l (median [IQR])	0.15 [0.09, 0.22]	0.16 [0.10, 0.23]	<0.001
Cholesterol in Medium HDL, mmol/l (median [IQR])	0.49 [0.42, 0.58]	0.48 [0.41, 0.57]	0.005

Abbreviations: SD, standard deviation; IQR, interquartile range; MET, metabolic equivalent of task HbA1c, hemoglobin A1C; mRNFL, macular retinal nerve fiber layer; HDL, high-density lipoprotein; VLDL, very low-density lipoprotein

Characteristic	No Glaucoma	Glaucoma	P-value
Sample size (%)	57,578 (98.7)	780 (1.3)	
Sex - Male (%)	26,887 (46.7)	409 (52.4)	0.002
Age at recruitment (mean (SD))	57 (8.0)	60 (7.0)	<0.001
Ethnicity (%)			<0.001
White	55,034 (96)	690 (89)	
Asian	1,129 (2.0)	41 (5.3)	
Black	574 (1.0)	27 (3.5)	
Other	841 (1.5)	22 (2.8)	
Smoking Status (%)			0.67
Never	200 (0.3)	4 (0.5)	
Prefer not to answer	31,593 (54.9)	418 (53.6)	
Previous	20,276 (35.2)	287 (36.8)	
Current	5,509 (9.6)	71 (9.1)	
MET, minutes per week (mean (SD))	2,483 (2,450)	2,612 (2,562)	0.14
Body mass index kg/m ² (mean (SD))	27.4 (4.8)	27.8 (4.7)	0.021
HbA1c, mmol/mol (mean (SD))	35.9 (5.7)	38.1 (7.8)	<0.001
Spherical Equivalent, diopter (mean (SD))	-0.03 (2.1)	0.01 (1.9)	0.63
Intraocular pressure, mmHg (mean (SD))	15.4 (2.4)	17.1 (3.9)	<0.001
mRNFL thickness, μ m (mean (SD))	28.7 (1.7)	28.6 (1.3)	0.02
Beta blocker use (%)	4,322 (7.5)	71 (9.1)	0.11
Caffeine intake, mg/day (mean (SD))	1,66.3 (67.4)	166.6 (61.5)	0.90
Alcohol intake, g/week (median [IQR])	84 [41.7, 149.3]	84 [42.5, 145.9]	0.91
Diabetes (%)	3209 (5.6)	104 (13.3)	<0.001
Oral steroid use (%)	1,538 (2.7)	39 (5.0)	<0.001
Coronary Artery Disease (%)	2,656 (4.6)	64 (8.2)	<0.001
Total cholesterol, mmol/l (median [IQR])	4.59 [3.98, 5.22]	4.49 [3.76, 5.13]	0.001
Lactate, mmol/l (median [IQR])	3.95 [3.24, 4.75]	3.84 [3.21, 4.69]	0.16
Concentration of Small HDL Particles, mmol/l (median [IQR])	0.0097 [0.0089, 0.011]	0.0095 [0.0088, 0.010]	0.002
Cholesteryl esters in small HDL, mmol/l (median [IQR])	0.33 [0.30, 0.36]	0.32 [0.30, 0.36]	0.001
Albumin, mmol/l (median [IQR])	39.4 [37.3, 41.45]	38.9 [36.7, 40.8]	<0.001
Total lipids in small HDL, mmol/l (median [IQR])	1.16 [1.06, 1.26]	1.15 [1.05, 1.25]	0.018
Citrate, mmol/l (median [IQR])	0.065 [0.057, 0.074]	0.065 [0.057, 0.073]	0.52
Pyruvate, mmol/l (median [IQR])	0.080 [0.06, 0.10]	0.079 [0.061, 0.098]	0.14
Alanine, mmol/l (median [IQR])	0.29 [0.24, 0.35]	0.29 [0.24, 0.35]	0.95
Phospholipids in Small HDL, mmol/l (median [IQR])	0.66 [0.60, 0.72]	0.66 [0.60, 0.71]	0.033

Abbreviations: SD, standard deviation; MET, metabolic equivalents; HbA1c, hemoglobin A1C; mRNFL, macula region retinal nerve fiber layer; HDL, high density lipoprotein; VLDL, very low density lipoprotein

Table 4.

Demographic and clinical characteristics in 2006-2010 of participants among the bottom 50% of glaucoma polygenic risk score (PRS).

Top 10% of glaucoma polygenic risk score			
Metabolites (Probit score)	Glaucoma	No Glaucoma	Adjusted P-value (NEF)
Lactate	-0.146	0.0239	8.8E-12
Pyruvate	-0.137	0.0117	2.9E-10
Citrate	-0.0693	0.0079	0.018
Triglycerides in Very Large VLDL	0.0606	-0.0043	0.10
Triglycerides in Chylomicrons and Extremely Large VLDL	0.0572	-0.0076	0.11
Acetoacetate	0.0744	0.0011	0.11
Cholesteryl Esters in Small HDL	-0.0239	0.0376	0.12
Cholesteryl Esters in Medium HDL	-0.0261	0.0276	0.13
Alanine	-0.0612	-0.0006	0.14
Triglycerides in Large VLDL	0.0629	0.0005	0.15

Bottom 50% of glaucoma polygenic risk score			
Metabolites (Probit score)	Glaucoma	No Glaucoma	Adjusted P-value (NEF)
Albumin	-0.110	-0.0099	0.047
Concentration of Small HDL Particles	-0.0943	-0.0062	0.071
Cholesteryl Esters in Small HDL	-0.0925	-0.0055	0.085
Cholesterol in Small HDL	-0.0919	-0.0063	0.088
Omega-3 Fatty Acids	-0.0896	-0.0052	0.092
Docosahexaenoic Acid	-0.0876	-0.0061	0.10
Total Lipids in Small HDL	-0.0844	-0.0051	0.12
Lactate	-0.0823	0.0031	0.12
Citrate	-0.0882	-0.0023	0.16
Alanine	-0.0766	0.0011	0.18

Potential confounders adjusted by regression include time since the last meal/drink (hours), age, age-squared, sex, ethnicity (Asian, Black, White, and other), season, time of day of specimen collection (morning, afternoon, night), smoking status (never, past, and current smoker), alcohol intake, caffeine intake, physical activity (metabolic equivalent of task [MET] hours/week), body mass index (kg/m²), average systolic blood pressure (mm Hg), history of diabetes (yes or no), HbA1c (mmol/mol), history of coronary artery disease, systemic beta-blocker use, oral steroid use, and spherical equivalent refractive error (diopters).

Table 5.

Metabolites associated with glaucoma among participants in the top decile and the bottom half of glaucoma polygenic risk score.

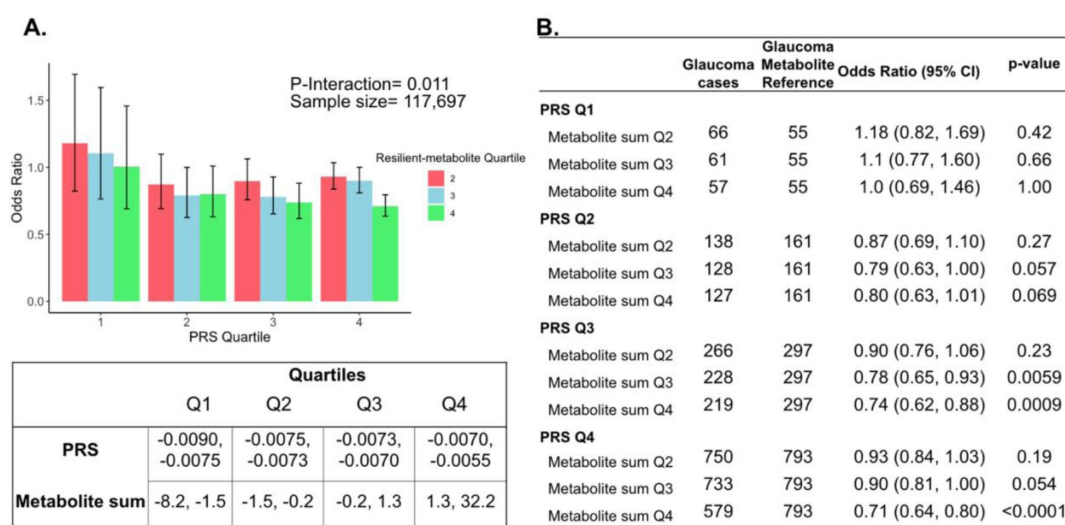


Figure 5.

Interaction of three putative resilient metabolites (lactate, pyruvate, and citrate) and polygenic risk score (PRS) on glaucoma risk.

(A) The bar chart shows the interaction of resilient probit-transformed metabolite sum with glaucoma genetic predisposition in each PRS quartile. In each glaucoma PRS quartile, the lowest metabolic sum quartile (Q1) is the metabolite reference group used to calculate the odds ratios. Each color represents resilient-metabolite sum quartiles (red = second quartile; blue = third quartile; and green = fourth quartile). Error bars show 95% confidence interval (CI). The table under the bar chart shows the ranges for the PRS and metabolite sum value quartiles. (B) The table shows odds ratios for glaucoma by PRS and putative resilient metabolite sum within various quartiles. The number of glaucoma cases within each resilient metabolite sum quartile and the number of glaucoma cases in the first quartile of resilient metabolite sum (Q1, labeled as glaucoma metabolite reference) are used to calculate the odd ratios. This analysis is adjusted for time since the last meal/drink (hours), age, age-squared, sex, ethnicity (Asian, Black, White, and other), season, time of day of specimen collection (morning, afternoon, night), smoking status (never, past, and current smoker), alcohol intake, caffeine intake, physical activity (metabolic equivalent of task [MET] hours/week), body mass index (kg/m²), average systolic blood pressure (mm Hg), history of diabetes (yes or no), HbA1c (mmol/mol), history of coronary artery disease, systemic beta-blocker use, oral steroid use, and spherical equivalent refractive error (diopters).

To confirm and visualize this interaction, we calculated and plotted the glaucoma odds ratio (OR) as a function of PRS quartile with each genetic predisposition bin stratified by the sum of the resilience-metabolite probit score quartiles (**Figures 5A** and **5B**). Notably, high levels of resilience-associated metabolites were significantly associated with lower odds of glaucoma, particularly within the higher PRS quartiles (Q3 and Q4). For example, for individuals in PRS Q3 with total resilience-associated metabolite sum Q3, the glaucoma OR was 0.78 (95% CI=0.65-0.93; $P=0.0059$), using metabolite sum Q1 as the reference. In the same PRS Q3 with a higher quartile of total resilience-associated metabolite sum (Q4), the OR was 0.74 (95% CI=0.62-0.88; $P=0.0009$). The lowest OR of glaucoma was observed in participants with both the highest PRS (Q4) and highest resilience-associated metabolite sum (Q4), with an OR of 0.71 (95% CI=0.64-0.80; $P<0.001$). Although both age and PRS were significant predictors of glaucoma prevalence (both are strongly associated with increased risk), there was no evidence of a significant three-way interaction between the resilience-associated metabolite sum, PRS, and age ($P_{3\text{-way-interaction}}=0.65$).

Next, we investigated whether a holistic metabolic risk score (MRS) incorporating resilience metabolites and other measured metabolites (full panel of 168 metabolites; see Methods and **Supplemental Table 3** for the beta coefficients used in MRS construction) could be used in conjunction with a glaucoma PRS to improve risk prediction. To quantify the degree of interaction of glaucoma PRS and MRS, we plotted glaucoma OR at various quartiles of glaucoma PRS and MRS compared to the first MRS quartile within each PRS quartile. We found a significant interaction between glaucoma MRS built from 168 metabolites and PRS ($P_{\text{interaction}}=0.0012$; see **Supplemental Figure 1A**).

In stratified analyses by quartile of PRS, compared to the first quartile of MRS, higher MRS was associated with increased glaucoma prevalence across most PRS quartiles, with the most pronounced effect observed in individuals in the highest PRS and MRS quartile with an OR of 2.14 compared to those in the highest PRS and lowest MRS quartile (**Supplemental Figure 1B**). Among individuals in the highest MRS quartile (Q4), PRS Q1 had an OR of 1.66 (95% CI=1.15-2.42; $P=0.0086$), PRS Q2 had an OR of 2.08 (95% CI=1.62-2.67; $P<0.001$), PRS Q3 had an OR of 2.20 (95% CI=1.83-2.65; $P<0.001$), and PRS Q4 had an OR of 2.14 (95% CI=1.91-2.40; $P<0.001$). This suggests potential synergistic effects of genetic and metabolite risk factors on glaucoma risk that transcend the impacts of the resilience metabolites.

Most importantly, using individuals in the lowest quartiles of PRS and MRS (PRS Q1, MRS Q1) as the reference group for the entire population ($n=117,698$), we determined glaucoma OR as a function of PRS quartile, with further stratification by holistic MRS in each PRS bin (**Figure 6**). The OR for PRS Q1 with MRS Q4 relative to PRS Q1 with MRS Q1 was modest (OR=1.66; 95% CI=1.15-2.42; $P=0.0086$). However, those with PRS Q4 (highest genetic risk) and MRS Q1 (lowest metabolic risk) had an OR of 11.7 (95% CI=8.72-16.0; $P<0.001$) compared to PRS Q1 MRS Q1. This illustrates the importance of genetics in impacting glaucoma risk. Strikingly, the glaucoma OR further increased to 25.1 (95% CI=18.8-34.1; $P<0.001$) for those with PRS Q4, MRS Q4 (combination of highest genetic and highest metabolic risk; $P_{\text{interaction}}=0.019$). Altogether, this demonstrates the strong prognostic utility of combining both PRS and MRS measurements for assessing glaucoma risk, particularly in individuals with a high genetic predisposition.

Pyruvate supplementation lessens intraocular pressure and glaucoma

To functionally test the association between higher levels of pyruvate and resilience to glaucoma, we experimentally tested whether treatment with pyruvate induces resilience to IOP elevation and glaucoma in a human-relevant mouse model. Common variants in *LMX1B* are associated with IOP variation and the most common form of human glaucoma, primary open-angle glaucoma (POAG).^{24,29} Rare Mendelian variants in *LMX1B* can produce early-onset ocular hypertension

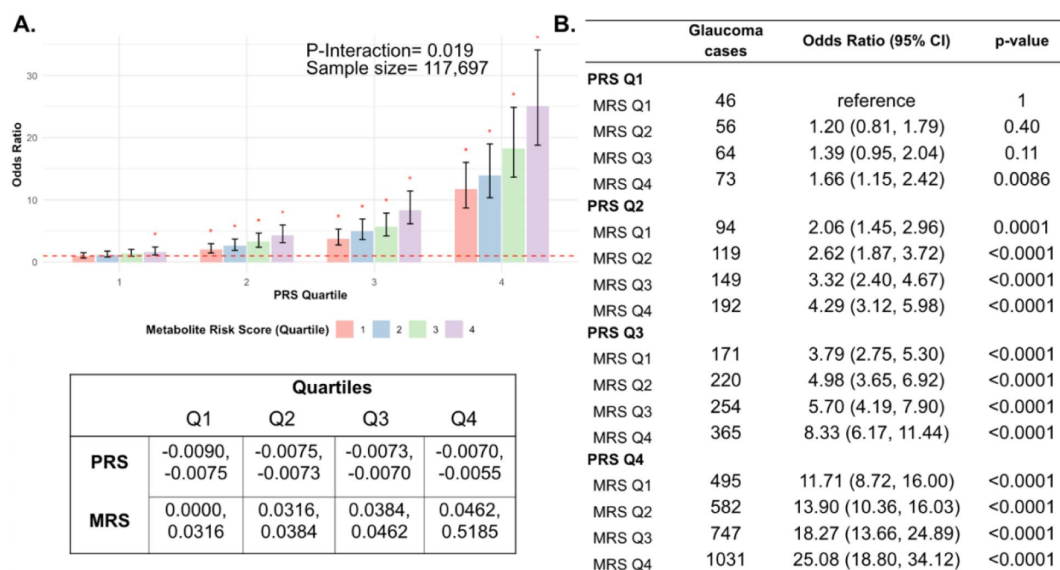


Figure 6.

Interaction of the holistic metabolite risk score (n=168 metabolites) and polygenic risk score (PRS) on glaucoma risk.

(A) The bar chart plots the odds ratio of glaucoma as a function of holistic probit-transformed MRS quartile with further stratification by glaucoma PRS in each MRS bin. The lowest quartile of glaucoma PRS and MRS is the reference group (see dotted red line) for the entire population. Each color represents the MRS quartiles (red = first quartile; blue = second quartile; green = third quartile; and purple = fourth quartile). Error bars show 95% confidence interval (CI). The table under the bar chart shows the ranges for the PRS and MRS quartiles. (B) Table showing odds ratios for glaucoma by polygenic risk score (PRS) and MRS within various quartiles. The number of glaucoma cases within each MRS and the number of glaucoma cases in PRS Q1 and MRS Q1 are used to calculate the odds ratios. This analysis is adjusted for time since the last meal/drink (hours), age, age-squared, sex, ethnicity (Asian, Black, White, and other), season, time of day of specimen collection (morning, afternoon, night), smoking status (never, past, and current smoker), alcohol intake, caffeine intake, physical activity (metabolic equivalent of task [MET] hours/week), body mass index (kg/m²), average systolic blood pressure (mm Hg), history of diabetes (yes or no), HbA1c (mmol/mol), history of coronary artery disease, systemic beta-blocker use, oral steroid use, and spherical equivalent refractive error (diopters).

and open-angle glaucoma (OAG).³⁰ We have previously demonstrated that mice with a dominant mutation in *Lmx1b* (*Lmx1b*^{V265D/+}) develop IOP elevation and glaucoma.^{30,31} Depending on the genetic background, this *Lmx1b*^{V265D} mutation induces either early-onset or later glaucoma.³¹ Mutant mice with a C57BL/6J strain background develop severe, early-onset IOP elevation and glaucoma. As pyruvate and its metabolites were associated with no glaucoma despite strong genetic predisposition in the UKBB cohort (highest decile of PRS), we tested the ability of dietary pyruvate to induce resilience against the *Lmx1b*^{V265D}-induced glaucoma on this C57BL/6J genetic background. Pyruvate supplementation through drinking water substantially protected mice from IOP elevation and glaucoma. Pyruvate significantly protected against both anterior chamber deepening (ACD), a consequence of IOP elevation in mouse eyes (**Figure 7A-B**), and IOP elevation itself (**Figure 7C**). Importantly, pyruvate treatment protected against glaucomatous optic nerve degeneration (**Figure 7D**). Together, our findings strongly support the role of endogenous pyruvate in conferring resilience against glaucoma even countering strong genetic predisposition. Further, they show that pyruvate can act as a potent resilience factor against IOP elevation and glaucoma when delivered orally.

Discussion

In this large cohort study, we found that the inclusion of metabolite data from an NMR platform only marginally improved various glaucoma prediction algorithms (**Figure 3**). Nonetheless, using an agnostic approach, we found that lactate, pyruvate, and citrate levels collectively were associated with a 29% reduced risk of glaucoma among those in the top quartile of glaucoma genetic predisposition (**Figure 5B**). We showed that pyruvate supplementation reduced glaucoma incidence in a human-relevant genetic mouse model (**Figure 7**). Furthermore, the interaction between glaucoma PRS and an MRS based on 168 metabolites provided synergistic predictive information regarding glaucoma risk (**Figure 6**). The statistical interaction we demonstrate between metabolite scores and PRS-derived glaucoma risk helps identify individuals who, despite high genetic risk, are less likely to develop glaucoma due to favorable metabolic profiles or, on the contrary, are more likely to develop glaucoma and should be monitored at an earlier age. Overall, these findings indicate that an MRS may provide clinically useful measures of metabolic status and improve patient risk stratification in glaucoma.

While earlier studies^{32–34} showed modest discriminatory powers for a glaucoma PRS, recent larger genome-wide association studies reported significantly improved risk prediction.^{7,35–37} However, no studies have examined the utility of incorporating metabolite data into glaucoma prediction algorithms combined with a PRS. Future work is needed to understand how non-genetic factors, known to alter glaucoma risk or IOP levels, like air pollution, psychological stress, physical activity, and dietary factors,¹⁴ could change the MRS and alter high glaucoma genetic predisposition.

Previous studies have explored metabolomics to identify biomarkers for OAG. A systematic review analyzing 13 studies identified a total of 144 metabolites, of which 12 were associated with OAG.²² Among studies using plasma samples, four metabolic pathways were significantly enriched: sphingolipid metabolism, arginine and proline metabolism, and beta-alanine metabolism. In another study across three US cohorts and the UKBB, higher levels of plasma diglycerides and triglycerides were adversely associated with glaucoma.³⁸ This is consistent with our observation that lower triglyceride levels were associated with glaucoma resilience, albeit with less significance compared to the three glycolysis-related metabolites. The biological mechanism is suggested to be the association of hypertriglyceridemia with increased blood viscosity and elevated IOP.^{39–42} Consistent with this, a meta-analysis reported that individuals with glaucoma had a mean triglyceride level 14.2 mg/dl higher than those without the disease.⁴¹

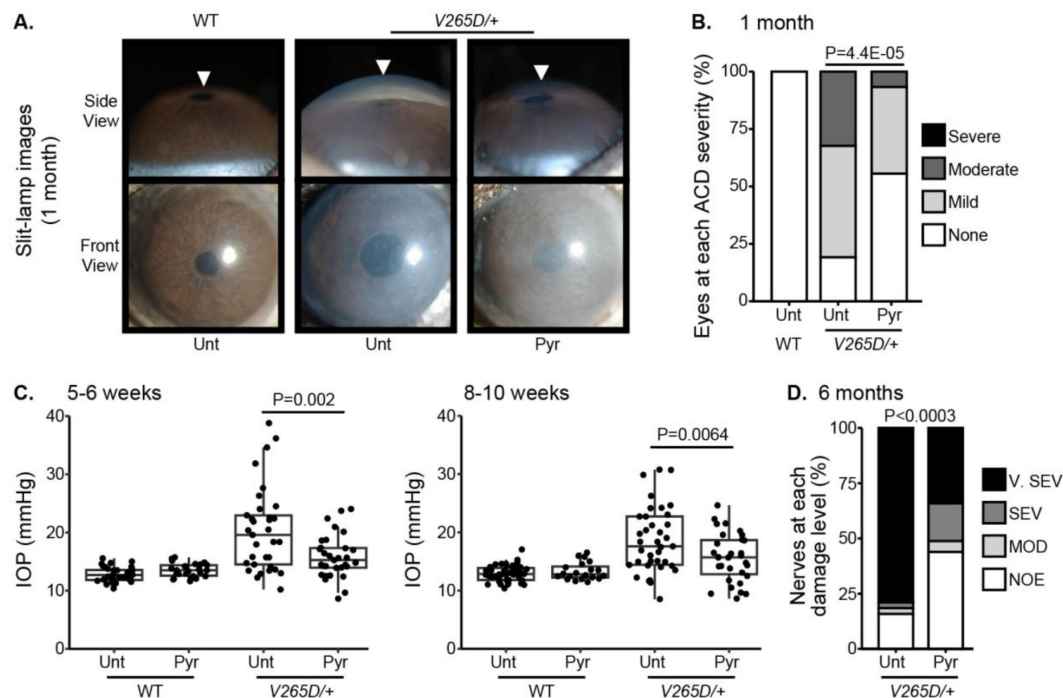


Figure 7.

Pyruvate treatment protects from IOP elevation and glaucoma.

(A) Representative photos of eyes from mice of the indicated genotypes and treatments (Unt = untreated, Pyr = pyruvate treated). *Lmx1b* is expressed in the iris and cornea, so *Lmx1b*^{V265D} mutant eyes have primary abnormalities of the iris and cornea. This includes corneal haze, which is present before IOP elevation in many eyes and likely reflects a direct transcriptional role of LMX1B in collagen gene expression. *Lmx1b*^{V265D} mutant eyes also develop anterior chamber deepening (ACD), a sensitive indicator of IOP elevation in mice. The WT and pyruvate-treated mutant eyes have shallow anterior chambers, while the untreated mutant eye has a deepened chamber (arrowheads). (B) Distributions of ACD are based on a previously defined scoring system.³¹ Groups are compared by Fisher's exact test. n > 30 eyes were examined in each group. (C) Boxplots of IOP (interquartile range and median line) in WT and mutant eyes. Pyruvate treatment significantly lessens IOP elevation in mutants compared to untreated mutant controls. Groups were compared by ANOVA followed by Tukey's honestly significant difference. n > 30 eyes were examined in each *Lmx1b*^{V265D} mutant group, and n > 20 eyes were examined in WT groups. (D) Distributions of damage based on analysis of PPD-stained optic nerve cross sections from 6-month-old mice (Methods). Pyruvate treatment lessened the incidence of glaucoma (Fisher's exact test). No glaucoma was found in WT mice. Geno = genotype. n > 40 nerves examined per group. NOE = no glaucoma, MOD = moderate. SEV = severe, and V. SEV = very severe (see Methods).

While studies have identified biomarkers associated with glaucoma, to the best of our knowledge, there has been no investigation on biomarkers that are associated with being resilient to a high glaucoma genetic predisposition. We identified elevated levels of lactate, pyruvate, and citrate among individuals without glaucoma despite high genetic risk. The retina commonly utilizes energy produced from glycolysis and oxidative phosphorylation. Therefore, glycolysis end products (pyruvate and lactate) are important energy sources for retinal ganglion cells (RGCs).⁴³ RGCs express monocarboxylate transporters, which facilitate the uptake of extracellular pyruvate and lactate, improving energy homeostasis, neuronal metabolism, and survival.^{44,45} Lactate is critical for RGC survival during periods of glucose deprivation,^{46,47} and bioenergetic insufficiency contributes to RGC loss in glaucoma.^{48–51} Finally, a study found that baseline lactate levels were significantly decreased in patients with normal-tension glaucoma compared to non-glaucomatous controls.⁵²

Pyruvate plays a central role in energy metabolism by linking glycolysis to the TCA cycle, ultimately driving ATP production by the electron transport chain. Pyruvate is converted into acetyl-CoA, which fuels the TCA cycle, and into lactate, a key molecule that regulates metabolism and serves as an important energy source.⁵³ The TCA cycle is tightly regulated to balance biosynthesis (producing metabolic precursors for lipids, carbohydrates, amino acids, nucleic acids, and co-factors) and energy production, adjusting dynamically to cellular needs.⁵⁴ Beyond its metabolic functions, pyruvate also acts as an antioxidant, scavenging reactive oxygen species (ROS), inducing the expression of the antioxidant response control gene *Nrf2*, and promoting beneficial autophagy.^{55–57} Its neuroprotective effects on RGCs have been demonstrated both in cell culture and in an induced model of glaucoma in rats.⁵⁸ Additionally, pyruvate combined with nicotinamide has been shown to enhance optic nerve protection in mice.⁵⁹ Importantly, the current study extends this evidence by demonstrating that pyruvate alone provides strong protection against glaucoma in a mouse model carrying a mutation in the ortholog of a human POAG gene. Further, it extends pyruvate's protective effects to the ocular drainage tissues, mitigating IOP elevation. Notably, although our validation data demonstrates the neuroprotective effects of exogenous pyruvate, it remains possible that endogenously produced pyruvate within ocular tissues may also contribute to RGC protection. A randomized Phase II clinical trial that tested pyruvate and nicotinamide in POAG patients found improved visual function in glaucoma patients compared to the placebo group.⁶⁰ Together, both preclinical and clinical evidence support the role of pyruvate as a protective factor against glaucoma.

Citrate is a key intermediate in the TCA cycle and is important for aerobic energy production. Elevated citrate levels might reflect enhanced energy supply and antioxidative capacity of ocular tissues, potentially protecting RGCs against oxidative stress.^{50,61–64} Decreased levels of citrate may indicate increased usage or mitochondrial dysfunction.⁶⁵ Studies have found lower plasma citrate levels in glaucoma patients among adults and children.^{66,67} A study measuring plasma citrate with a cutoff of 110 $\mu\text{mol/L}$ to detect glaucoma had a sensitivity of 66.7% and a specificity of 71.4%.⁶⁶ These findings corroborate the protective effects of citrate in conferring resilience to glaucoma. Similar to pyruvate, plasma citrate may also play a role in retinal homeostasis, highlighting the need to disentangle serum-derived from intraocularly synthesized metabolites.

We also identified several lipoproteins associated with resilience, most notably cholesteryl esters in small high-density lipoprotein (HDL) and medium HDL. Although these lipoproteins did not reach an adjusted $P\text{-value} < 0.05$ and thus remain suggestive biomarkers rather than definitive, this is consistent with studies describing the neuroprotective effect of HDL in the glaucoma pathophysiology.^{68,69} Cholesteryl esters and triglycerides make up the core of lipoproteins, which transport lipids in the blood. Lipoproteins are characterized by their size, where HDL ranges from 8–12 nm in diameter, and low-density lipoprotein (LDL) ranges from 18–25 nm in diameter. The retinal pigment epithelium can take up lipoproteins,^{70,71} and dysregulation of lipid metabolism can result in the accumulation of lipids and oxidative stress in the retina,

resulting in RNFL thinning^{38,72–75}. Thus, HDL has antioxidant properties to protect RGCs by facilitating cholesterol efflux from the retina^{76,77}. Furthermore, as HDL analytes can cross blood-ocular barriers⁷⁸, there is a plausible route for serum-derived HDL to influence RGC homeostasis. Nonetheless, the relative contributions of circulating lipoproteins versus local synthesis within ocular tissues remain unclear and warrant further investigation. Notably, genes implicated by large cross-ancestry and European subset GWAS meta-analyses of POAG risk are enriched in processes related to apolipoprotein binding and lipid transport^{26,78}. Incorporating ocular physiology and blood-retinal barrier considerations when interpreting lipoproteins as potential resilience biomarkers will be critical for future studies aimed at understanding and therapeutically targeting increased glaucoma risk.

This study had several strengths. First, the UKBB is a large cohort study with detailed covariate information on demographics, clinical characteristics, and glaucoma endophenotypes, allowing for better glaucoma risk profiling. Second, we used a PRS calculated from the large (~600k individuals) multi-trait analysis of genome-wide association studies on glaucoma and endophenotypes, including optic nerve head structural features and IOP data⁷. Finally, we supported our biobank analyses with functional tests in a disease-relevant mouse model, identifying a protective effect of pyruvate supplementation in animals susceptible to glaucoma. The data showing that pyruvate protected against IOP elevation (a key causative risk factor for glaucoma) and glaucomatous nerve damage in a mouse model, as well as pyruvate's biochemical relatedness to lactate and citrate regarding energy metabolism, mitigate against concerns about false discovery, uncontrolled confounding, reverse causation, and collinearity among biomarkers.

A major limitation of our study is that it includes a predominately European population, with an underrepresentation of African, Asian, and mixed American genetic ancestral groups; nonetheless, we chose to analyze Europeans and non-Europeans together to make the results as generalizable as possible. A second limitation is that glaucoma ascertainment in the UK Biobank is based on self-reported diagnoses and hospital records rather than comprehensive ophthalmologic examinations. Nonetheless, it is reassuring that the prevalence of glaucoma in our sample (~4%) is similar to a directly performed disease burden estimate in a comparable, albeit slightly older, United Kingdom sample (2.7%)⁷⁹. In addition, a subset of UKBB participants have received glaucoma treatment, which could alter metabolite levels and could lead to an underestimate of the protective effects of metabolites. Moreover, the UKBB study includes a limited set of 168 metabolites, which is not inclusive of the comprehensive set of metabolites, known and unknown. This study also only focuses on plasma metabolites and does not account for ocular metabolite profiles. Due to the cross-sectional design of our study, establishing temporality is challenging. As a result, it remains unclear whether the observed metabolic changes contribute to the onset of glaucoma or arise as a consequence of the disease. However, the mouse data support a causal role for pyruvate as a factor that mitigates high genetic risk. This study's exploratory approach—investigating metabolite associations in an agnostic, hypothesis-free manner—underscores the need for replication. Regarding the mouse studies, we performed ad-lib pyruvate dosing and examined only a single human-related mutation, whereas individuals with a genetic predisposition to glaucoma typically carry a high cumulative burden of multiple common glaucoma-associated variants. Lastly, neither the mouse studies, nor the human data accurately inform the exact dosing of TCA cycle biomarkers needed to mitigate glaucoma risk. We find considerable overlap between TCA cycle biomarkers between those with and without glaucoma in the top decile of glaucoma risk (Supplementary Table 1² and Supplementary Figure 2²).

Overall, we have shown that plasma metabolites marginally enhance the predictive power of a glaucoma PRS and may have a role in identifying those who may be resilient to glaucoma despite high genetic predisposition. Our study identified protective metabolites linked to glycolysis and mitochondrial function, suggesting important pathophysiology in glaucoma. One of these metabolites of significance is pyruvate. Oral pyruvate supplementation substantially protected

against IOP elevation and conferred resilience against glaucoma. Thus, these results open new avenues for therapeutic strategies of pyruvate and its metabolites as resilience factors against glaucoma in genetically predisposed individuals.

Methods

Study Design

Our initial human analysis was conducted in three steps. First, we assessed whether plasma metabolite data alone can predict glaucoma risk with receiver operating characteristic (ROC) curves in the UK Biobank. We then integrated the metabolite data into a polygenic risk score (PRS)-based glaucoma risk assessment model to see if metabolites could enhance glaucoma prediction. Second, we focused on identifying a metabolomic signature of resilience to high glaucoma PRS by comparing plasma metabolites in glaucoma cases versus participants without glaucoma (henceforth referred to as resilient participants) from the top 10% of the glaucoma PRS. Third, we explored the interactions between metabolic scores and PRS in modifying glaucoma risk.

Study population and data collection

The UKBB is a prospective cohort study of over half a million participants aged 37 to 73 years at recruitment across the United Kingdom from 2006 to 2010. Participants were recruited through the National Health Service registers from 22 assessment centers, where they signed electronic informed consent to participate. Participants then completed in-depth touchscreen questionnaires and trained staff-led interviews, performed body measurements, and provided biological samples.⁸⁰ Additional outcomes are available from data linkage to hospital episode statistics, the death register, and primary data.^{80–82} Biological samples collected from these participants, including blood, urine, and saliva specimens, were used to generate genetic, metabolomic, and proteomic data.⁸³ Details of the UKBB study design and population are described online (<https://www.ukbiobank.ac.uk>). The UKBB study received approval from the National Health Service North West Multicentre Research Ethics Committee (reference number 06/MRE08/65) and the National Information Governance Board for Health and Social Care. This research, conducted under UK Biobank application number 36741, adhered to the tenets of the Declaration of Helsinki.

The UKBB included 502,613 participants, of which 173,679 participants had self-reported glaucoma data, ICD-coded, or previous glaucoma laser/surgical therapy (34.6%) at baseline. A subset consisting of 117,698 participants with metabolomic data and genetic profiling comprise the study population (**Figure 1**). The glaucoma cases were defined based on those with self-reported glaucoma, in which participants selected “glaucoma” when they completed a touchscreen questionnaire with the question, “Has a doctor told you that you have any of the following problems with your eyes?”. Glaucoma cases also included participants who reported a history of glaucoma surgery or laser therapy on the questionnaire or if an International Classification of Diseases (ICD) code for glaucoma (ICD 9th revision: 365.* [excluding 365.0]; ICD 10th revision: H40.* [excluding H40.0] and H42.*) was carried in linked Hospital Episode Statistics before the baseline assessment. The described approach to identifying glaucoma has been strongly supported by prior research publications.^{27,28}

Metabolite Profiling

In the UKBB, a high-throughput nuclear magnetic resonance (NMR)-based biomarker platform (Nightingale Health Ltd; Helsinki, Finland) was used to measure the metabolomic profile in the randomly selected non-fasting EDTA plasma samples from a subset of participants.⁸⁴ In contrast to liquid chromatography/mass spectroscopy, NMR spectroscopy produces distinctive spectral shapes for molecules containing hydrogen atoms.⁸⁵ The areas under the curve (AUC) are proportional to the concentration of each molecule based on chemical shifts and J coupling split

patterns derived from quantum mechanics.⁸⁶⁸⁷ The NMR platform contains data on 249 metabolic biomarkers (168 absolute levels and 81 ratio measures), including a subset of 36 biomarkers (27 absolute levels and 9 ratio measures) certified for broad diagnostic use by the European Union (EU).⁸⁴ In this study, we focused on the 168-metabolite and the 27-metabolite sets for analysis (for a complete list of quantified metabolites, refer to the Supplementary File). Individual metabolite values were transformed to probit scores to standardize their range and reduce the impact of skewed distributions.

Construction of PRS for glaucoma

The glaucoma PRS used in this study was from a multi-trait analysis of GWAS (MTAG) on glaucoma.⁷ This PRS was derived based on GWAS data from glaucoma (7,947 cases and 119,318 controls) and its endophenotypes: optic nerve head structure using vertical cup-disc ratio (VCDR) (including additional data from 67,040 UKBB and 23,899 International Glaucoma Genetics Consortium (IGGC) participants), and intraocular pressure (IOP, including additional data on 103,914 UKBB and 29,578 IGGC participants). The glaucoma PRS was constructed using single nucleotide polymorphisms (SNPs) with MTAG P-values ≤ 0.001 , resulting in 2,673 uncorrelated SNPs after LD-clumping at $r^2 = 0.1$, and was found to have a predictive ability with an area under the curve (AUC) of 0.68 (95% confidence interval, CI, [0.67-0.70]) and 0.80 accounting for age, sex, family history, and PRS tested in the Australian and New Zealand Registry of Advanced Glaucoma cohort.

Model building/covariates

We first assessed whether the inclusion of probit-transformed metabolite data enhances glaucoma prediction algorithms. We adjusted for factors involving major determinants of variability in metabolites, POAG-established and suspected risk factors, and other comorbidities. In model 1, we considered metabolites only. Model 2 incorporated additional demographic covariates, including age (years), sex, genetic ancestry, season, time of day of specimen collection (morning, afternoon, night), and fasting time (hours). Model 3 incorporated covariates in model 2 and additional variables accounting for comorbidities potentially related to glaucoma including smoking status (never, past, and current smoker), alcohol intake (g/week),¹⁰ caffeine intake (mg/day),⁸⁸ physical activity (metabolic equivalent of task [MET], hours/week), body mass index (kg/m^2), average systolic blood pressure (mm Hg), history of diabetes, hemoglobin A1C (HbA1c, mmol/mol), self-reported history of coronary artery disease, systemic beta-blocker use, oral steroid use, and mean spherical equivalent refractive error (diopters) across both eyes. Model 4 incorporated covariates in model 3 and a glaucoma PRS. We compared these models with and without the metabolites to determine the predictive ability of the metabolites above and beyond a glaucoma PRS and known risk factors. Missing values were imputed with medians for numeric variables and modes for factor variables.

Statistical analysis

Metabolite-Based Predictive Modeling

Multivariable logistic regression models with regularization (implemented with *glmnet*) were built to investigate the associations of metabolic biomarkers and the risk of glaucoma. We used four sequential models, adjusting for covariates as described above. For each model, we examined three groupings: no metabolites, the 27 metabolites corresponding to EU-stamped validated markers, and 168 metabolites corresponding to all the measured plasma metabolites captured in the UKBB. We used regularization to address collinearity, which reduced the set of metabolites and other covariates considered in each model (reported in [Supplementary Table 2](#)). To examine model accuracy for predicting glaucoma prevalence, we utilized ROC curves and associated AUC measurements as a metric for model performance. We performed five-fold cross-validation and split data into 80% used for training and 20% used for evaluation. All metabolites' values were

probit transformed and used as continuous variables (per 1 standard deviation (SD) increase). Additionally, we stratified glaucoma by age, ethnicity, and sex. The DeLong test was utilized to examine the statistical significance of AUC differences between models, with a threshold of $P < 0.05$ as statistically significant.

Identifying resilience metabolites to high glaucoma polygenic risk scores

To identify individual metabolites associated with resilience to a high glaucoma PRS, we first performed a logistic regression model to obtain metabolite residuals from probit-transformed concentration data adjusting for the following variables: age, age-squared, time since the last meal/drink (≤ 4 , 5–8, and 9+ hours), sex, ethnicity (Asian, Black, White, and mixed/other), season, time of day of specimen collection, smoking status (never, past, and current smoker), alcohol intake, caffeine intake, physical activity (metabolic equivalent of task [MET] hours/week), body mass index (kg/m^2), average systolic blood pressure (mm Hg), history of diabetes, HbA1c (mmol/mol), history of coronary artery disease, systemic beta-blocker use, oral steroid use, and spherical equivalent refractive error (diopters). Metabolites associated with glaucoma were nominated by performing a t-test comparing residuals in participants with and without glaucoma to identify metabolites significantly associated with glaucoma in both the top 10% and the bottom 50% of glaucoma PRS score. This stratification balanced case counts to enable comparison of groups with adequate sample size (780 glaucoma cases in the bottom 50% of PRS score and 1,693 glaucoma cases in the top 10% of PRS score). To account for covariance in metabolite abundance, we corrected for multiple comparisons using the number of effective tests (NEF) method.⁸⁹ First, a correlation matrix was generated using pairwise complete observations to address any missing values. Eigenvalue decomposition was performed on the correlation matrix to capture the principal components. Principal components contributing more than 1% of the total variance were calculated, yielding 9 significant components, capturing 91.6% of the variance. Thus, we applied a Bonferroni adjustment based on the number of significant components ($n=9$) to calculate the adjusted p-values accounting for multiple hypotheses. P-values are considered statistically significant if the NEF-adjusted P-value was < 0.05 .⁸⁹ Given this was an exploratory analysis, NEF-adjust P-value < 0.2 were considered worthy of additional analysis.

Construction of Metabolite Risk Scores

We constructed metabolite risk scores (MRS), which were calculated as a weighted sum of each metabolic biomarker from coefficients in the 168-metabolite base model. Specifically, we utilized a logistic regression model (implemented with a *binomial glm*) using probit transformed metabolite values to predict the prevalence of glaucoma using five-fold cross-validation. The resulting beta values were extracted and are reported in [Supplementary Table 3](#). To determine if there is an interaction between glaucoma MRS and PRS, we utilized extracted model coefficients and corresponding significance from a binomial generalized linear model. We fit this model using the formula $\text{Glaucoma} \sim \text{PRS} * \text{MRS} + \text{PRS} + \text{MRS} + \text{Covariates}$. We classified participants into 16 groups by quartiles of MRS and PRS. Within each PRS quartile, we used MRS quartile 1 as the reference for comparison. Additionally, we used the group in both PRS quartile 1 and MRS quartile 1 as the overall reference group for all comparisons among the groups. A chi-squared test was used to assess statistical significance between odds ratios (ORs) with 95% confidence intervals (CIs).

To investigate whether age modifies the associations of resilience metabolites with glaucoma, we conducted a stratified analysis and assessed a three-way interaction term of age ($\geq$ or < 58 years old, based on median age), the sum of resilience metabolite levels, and PRS using a Wald test for individuals in the top 10% PRS. All statistical analyses and plots were produced using R version 4.2.1 (R Foundation for Statistical Computing, Vienna, Austria). All statistical tests were two-sided.

Animal husbandry and ethics statement

Experimental mice had a C57BL/6J (Jackson Laboratory Stock #000664) genetic background. The *Lmx1b*^{V265D} (alias *Lmx1b*^{Icst}) mutation is previously reported to cause high IOP and glaucoma in mice, with IOP becoming elevated in some eyes during the first months after birth.^{30,31,90} The mutation was backcrossed to the C57BL/6J strain for at least 30 generations. All mice were treated per the Association for Research in Vision and Ophthalmology's statement on the use of animals in ophthalmic research. The Institutional Animal Care and Use Committee of Columbia University approved all experimental protocols performed. Mice were maintained on PicoLab Rodent Diet 20 (5053, 4.5% fat) and provided with reverse osmosis-filtered water. Mutant and control littermates were housed together in cages containing ¼-inch corn cob bedding, covered with polyester filters. The animal facility was maintained at a constant temperature of 22°C with a 14-hour light and 10-hour dark cycle.

Genotyping mice

Lmx1b^{V265D} and *Lmx1b*⁺ genotypes were determined by direct Sanger sequencing of a specific PCR product. Genomic DNA was PCR amplified with forward primer 5'-CTTGAGCCATCGGAGCTG-3' and reverse primer 5'-ATCTCCGACCGCTTCCTGAT-3' using the following program: (1) 94 °C for 3 min; (2) 94 °C for 30 s; (3) 57 °C for 30 s; (4) 72 °C for 1 min; (5) repeat steps 2–4 35 times; and (6) 72 °C for 5 min. PCR products were purified and sequenced by the Genewiz (Azenta Life Sciences).

Slit-lamp examination

Anterior segments were examined approximately every 2 weeks between 1 month and 3 months of age and examined at monthly intervals between 3 months of age and harvest age (6 months). Balanced groups of males and females were examined as previously published.^{30,31} Photographs were taken with a 40× objective lens. Anterior chamber deepening, a sign of exposure to high IOP, was graded based on a semiquantitative scale of no deepening (not present), mild, moderate, or severe, as previously published.³¹ Under examination by either a slit-lamp or dissection scope (during IOP procedure), anterior chamber deepening was only detected in *Lmx1b* mutant groups. Groups were compared statistically by Fisher's exact test. $n > 40$ eyes were examined in each group.

Intraocular pressure measurement

IOP was measured with the microneedle method as previously described in detail.^{91,92} Before cannulation, mice were acclimatized to the procedure room and anesthetized via an intraperitoneal injection of a mixture of ketamine (99 mg/kg; Ketlar, Parke-Davis, Paramus, NJ, USA) and xylazine (9 mg/kg; Rompun, Phoenix Pharmaceutical, St Joseph, MO, USA) immediately prior to IOP assessment, a procedure that does not alter IOP in the experimental window.⁹² IOP was measured at both 5–6 weeks and 8–10 weeks of age in WT and *Lmx1b*^{V265D/+} eyes. Balanced groups of males and females were examined. During each IOP measurement period, the eyes of independent WT B6 mice were assessed in parallel with experimental mice as a methodological control to ensure proper calibration and equipment function. *Lmx1b* mutant groups were compared by ANOVA followed by Tukey's honestly significant difference. $n > 30$ eyes were examined in each *Lmx1b*^{V265D} mutant group and $n > 20$ eyes were examined in WT groups.

Pyruvate administration

Ethyl pyruvate (Sigma-Aldrich, St. Louis, Missouri) was dissolved in the standard institutional drinking water to a dose of 2000 mg/kg/day for adult mice based on the average volume mice consume. The mothers consumed this dose and delivered it to their suckling pups at an unknown dose throughout the first 3 to 4 weeks of life. At 4 weeks of age, the mice were weaned into

separate cages. From 4 weeks on, the young mice consumed a dose of approximately 2000 mg/kg/day based on their water consumption and body weight over weekly intervals. Untreated groups received the same drinking water without pyruvate. The water was changed once per week. Treatment was started on postnatal day 2. Births were checked daily between 9 am and 12 pm to determine the pups' age.

Optic nerve assessment

We analyzed *Lmx1b*^{V265D/+} and WT controls optic nerve for glaucomatous damage at 6 months of age (5.3-5.9 month range, majority are 5.8-5.9 months, sex balanced). Intracranial portions of optic nerves were dissected, processed, and analyzed as previously described.^{48,93,94} Briefly, optic nerve cross-sections were stained with para-phenylenediamine (PPD) and examined for glaucomatous damage. PPD stains all myelin sheaths, but differentially darkly stains the myelin sheaths and the axoplasm of damaged axons. This allows for the sensitive detection and quantification of axon damage and loss. Optic nerves were prepared for analysis with a 48-h fixation in 0.8% paraformaldehyde and 1.2% glutaraldehyde in 0.08 M phosphate buffer (pH 7.4) at 4 °C followed by overnight post-fix in osmium tetroxide at 4 °C. Nerves were washed twice for 10 min in 0.1 M phosphate buffer, once in 0.1 M sodium-acetate buffer, and dehydrated in graded ethanol concentrations. Tissues were then embedded in Embed 812 resin (Electron Microscopy Sciences, Fort Washington, PA, USA), and 1 µm-thick sections were stained in 1% PPD for ~40 minutes. Stained sections were compared using a damage scale that is validated against axon counting.^{93,95} Multiple sections of each nerve were considered when determining damage level. Nerves were determined to have one of 4 damage levels: 1) No glaucoma (NOE) – less than 5% axons damaged. This level of damage is seen in age- and sex-matched non-glaucomatous mice and is not due to glaucoma. We named this level no or early stage as some have early molecular changes when assessed with transcriptomics, but they cannot be distinguished from control by morphology.^{50,96} 2) Moderate damage (MOD) – average of 30% axon loss; 3) Severe (SEV) – greater than 50% axonal loss and extensive axon damage; and 4) Very severe (V. SEV) – glial scar over the vast majority of nerve with few remaining axons.

Data availability

Data from the UK Biobank cannot be shared due to our Material Transfer Agreement. To request access to the UK Biobank data, please make requests directly to the UK Biobank via <https://www.ukbiobank.ac.uk/enable-your-research/apply-for-access>.

Acknowledgements

The authors would like to thank the participants who contributed their data to the UK Biobank study for this study. The authors would also like to thank the members of the Simon John Laboratory for experimental and technical assistance and the Institute of Comparative Medicine for animal and veterinary care.

Additional information

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UK Biobank Eye and Vision Consortium

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CRediT authorship contribution statement

Keva Li: Writing – original draft, Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. Nicholas Tolman: original draft, Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. Ayellet V. Segrè: Writing – review & editing, Methodology, Investigation. Kelsey V. Stuart: Data Curation, Methodology, Resources, Writing – review & editing, Methodology, Investigation. Oana A. Zeleznik: Writing – review & editing, Investigation. Neeru A. Vallabh: Writing – review & editing, Investigation. Kuang Hu: Writing – review & editing, Investigation. Nazlee Zebardast: Writing – review & editing, Investigation. Akiko Hanyuda: Writing – review & editing, Investigation. Yoshihiko Raita: Writing – review & editing, Investigation. Christa Montgomery: Writing – review & editing, Investigation. Chi Zhang: Writing – review & editing, Investigation. Pirro G. Hysi: Writing – review & editing, Resources, Project administration, Methodology, Investigation, Funding acquisition. Ron Do: Writing – review & editing, Investigation. Anthony Khawaja: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. Janey L. Wiggs: Writing – review & editing, Investigation, Funding acquisition. Jae H. Kang: Writing – review & editing, Supervision, Resources, Methodology, Investigation, Funding acquisition. Simon WM. John: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization, Funding acquisition. Louis R. Pasquale: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. All authors were involved in the interpretation of the results, and the critical review and approval of the manuscript.

Financial Support

This work is supported by NEI R01 015473 (LRP), NEI R01 032559 (LRP, JLW, AVS), EY036460 (LRP, JHK), NEI P30 EY014104 (JLW, AVS), NEI R01EY031424 (AVS), NIGMS R35-GM124836 (RD), NEI EY032507, EY032062, and EY018606 (SWMJ), an unrestricted Challenge Grant from Research to Prevent Blindness (NYC), and The Glaucoma Foundation (NYC). Further funding was provided by startup funds at Columbia University including the Precision Medicine Initiative, and by the New York Fund for Innovation in Research and Scientific Talent (NYFIRST; EMPIRE CU19-2660; SWMJ). This work is also supported by K23 (1K23EY032634) and the Research to Prevent Blindness Career Development Award (N.Z), a Vision Core grant P30EY019007 (Columbia University) an unrestricted departmental award from Research to Prevent Blindness (Columbia University). Supported by Fight for Sight (UK) (1956A), and The Desmond Foundation (K.V.S.); UK Research and Innovation Future Leaders Fellowship (MR/T040912/1), Alcon Research Institute Young Investigator Award and a Lister Institute of Preventative Medicine Fellowship (A.P.K.); financial support from the UK Department of Health through an award made by the National Institute for Health Research (NIHR) to Moorfields Eye Hospital National Health Service (NHS) Foundation Trust and University College London (UCL) Institute of Ophthalmology for a Biomedical Research Centre (BRC) for Ophthalmology (A.P.K.). The sponsor or funding organizations had no role in the design or conduct of this research. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

Abbreviations and Acronyms

- AUC: area under the curve
- BMI: body mass index
- CI: confidence interval
- HDL: high-density lipoprotein
- ICD: International Classification of Diseases
- IOP: intraocular pressure
- mRNFL: macular retinal nerve fiber layer

- MET: metabolic equivalent of task
- MRS: metabolic risk score
- MTAG: Multi-Trait Analysis of Genome-wide association study data
- NEF: number of effective
- NMR: nuclear magnetic resonance
- OAG: open-angle glaucoma
- OR: odds ratio
- POAG: primary open-angle glaucoma
- PRS: polygenic risk score
- RGC: retinal ganglion cell
- ROC: receiver operating characteristic
- SD: standard deviation
- SNP: single nucleotide polymorphism
- TCA: Tricarboxylic acid
- ATP: Adenosine triphosphate
- VCDR: vertical cup-disc ratio
- VLDL: very low-density lipoprotein.

Additional files

Supplemental Excel [↗](#)

Supplemental File [↗](#)

References

1. Soh Z, Yu M, Betzler BK, et al. 2021) **The Global Extent of Undetected Glaucoma in Adults: A Systematic Review and Meta-analysis** *Ophthalmology* **128**:1393–1404 <https://doi.org/10.1016/j.ophtha.2021.04.009>
2. Khawaja AP, Viswanathan AC 2018) **Are we ready for genetic testing for primary open-angle glaucoma?** *Eye* **32**:877–883 <https://doi.org/10.1038/s41433-017-0011-1>
3. Jan C, He M, Vingrys A, Zhu Z, Stafford RS 2024) **Diagnosing glaucoma in primary eye care and the role of Artificial Intelligence applications for reducing the prevalence of undetected glaucoma in Australia** *Eye* **38**:2003–2013 <https://doi.org/10.1038/s41433-024-03026-z>
4. Moyer VA, U. S. Preventive Services Task Force 2013) **Screening for Glaucoma: U.S. Preventive Services Task Force Recommendation Statement** . *Ann Intern Med* **159**:484–489 <https://doi.org/10.7326/0003-4819-159-6-201309170-00686>
5. US Preventive Services Task Force 2022) **Screening for Primary Open-Angle Glaucoma: US Preventive Services Task Force Recommendation Statement** *JAMA* **327**:1992–1997 <https://doi.org/10.1001/jama.2022.7013>
6. Zukerman R, Harris A, Oddone F, Siesky B, Verticchio Vercellin A, Ciulla TA 2021) **Glaucoma Heritability: Molecular Mechanisms of Disease** *Genes* **12**:1135 <https://doi.org/10.3390/genes12081135>
7. Craig JE, Han X, Qassim A, et al. 2020) **Multitrait analysis of glaucoma identifies new risk loci and enables polygenic prediction of disease susceptibility and progression** *Nat Genet* **52**:160–166 <https://doi.org/10.1038/s41588-019-0556-y>
8. de Vries VA, Hanyuda A, Vergroesen JE, et al. 2024) **The Clinical Usefulness of a Glaucoma Polygenic Risk Score in 4 Population-Based European Ancestry Cohorts** *Ophthalmology*. Published online August **14** <https://doi.org/10.1016/j.ophtha.2024.08.005>
9. Sekimitsu S, Xiang D, Smith SL, et al. 2023) **Deep Ocular Phenotyping Across Primary Open-Angle Glaucoma Genetic Burden** *JAMA Ophthalmol* **141**:891–899 <https://doi.org/10.1001/jamaophthalmol.2023.3645>
10. Stuart KV, Luben RN, Warwick AN, et al. 2022) **The Association of Alcohol Consumption with Glaucoma and Related Traits: Findings from the UK Biobank** *Ophthalmol Glaucoma* **6**:366 <https://doi.org/10.1016/j.ogla.2022.11.008>
11. Kang JH, Willett WC, Rosner BA, Buys E, Wiggs JL, Pasquale LR 2016) **Association of Dietary Nitrate Intake With Primary Open-Angle Glaucoma: A Prospective Analysis From the Nurses' Health Study and Health Professionals Follow-up Study** *JAMA Ophthalmol* **134**:294–303 <https://doi.org/10.1001/jamaophthalmol.2015.5601>
12. Kang JH, Ivey KL, Boumenna T, Rosner B, Wiggs JL, Pasquale LR 2018) **Prospective study of flavonoid intake and risk of primary open-angle glaucoma** *Acta Ophthalmol (Copenh)* **96**:e692 <https://doi.org/10.1111/aos.13705>

13. Kang JH, Wu J, Cho E, et al. 2016) **Contribution of the Nurses' Health Study to the Epidemiology of Cataract, Age-Related Macular Degeneration, and Glaucoma** *Am J Public Health* **106**:1684 <https://doi.org/10.2105/AJPH.2016.303317>
14. Stuart KV, Pasquale LR, Kang JH, Foster PJ, Khawaja AP 2023) **Towards modifying the genetic predisposition for glaucoma: An overview of the contribution and interaction of genetic and environmental factors** *Mol Aspects Med* **93**:101203 <https://doi.org/10.1016/j.mam.2023.101203>
15. Doucette LP, Rasnitsyn A, Seifi M, Walter MA 2015) **The interactions of genes, age, and environment in glaucoma pathogenesis** *Surv Ophthalmol* **60**:310–326 <https://doi.org/10.1016/j.survophthal.2015.01.004>
16. Nazifova-Tasinova N, Radeva M, Galunska B, Grupcheva C 2020) **Metabolomic analysis in ophthalmology** *Biomed Pap* **164**:236–246 <https://doi.org/10.5507/bp.2020.028>
17. Johnson CH, Ivanisevic J, Siuzdak G 2016) **Metabolomics: beyond biomarkers and towards mechanisms** *Nat Rev Mol Cell Biol* **17**:451–459 <https://doi.org/10.1038/nrm.2016.25>
18. Javadiyan S, Burdon KP, Whiting MJ, et al. 2012) **Elevation of Serum Asymmetrical and Symmetrical Dimethylarginine in Patients with Advanced Glaucoma** *Invest Ophthalmol Vis Sci* **53**:1923–1927 <https://doi.org/10.1167/iovs.11-8420>
19. Burgess LG, Uppal K, Walker DI, et al. 2015) **Metabolome-Wide Association Study of Primary Open Angle Glaucoma** *Invest Ophthalmol Vis Sci* **56**:5020–5028 <https://doi.org/10.1167/iovs.15-16702>
20. Nzoughet J Kouassi, Guehlouz K, Leruez S, et al. 2020) **A Data Mining Metabolomics Exploration of Glaucoma** *Metabolites* **10**:49 <https://doi.org/10.3390/metabo10020049>
21. Leruez S, Marill A, Bresson T, et al. 2018) **A Metabolomics Profiling of Glaucoma Points to Mitochondrial Dysfunction, Senescence, and Polyamines Deficiency** *Invest Ophthalmol Vis Sci* **59**:4355–4361 <https://doi.org/10.1167/iovs.18-24938>
22. Wang Y, Hou XW, Liang G, Pan CW 2021) **Metabolomics in Glaucoma: A Systematic Review** *Invest Ophthalmol Vis Sci* **62**:9 <https://doi.org/10.1167/iovs.62.6.9>
23. Privé F, Aschard H, Carmi S, et al. 2022) **Portability of 245 polygenic scores when derived from the UK Biobank and applied to 9 ancestry groups from the same cohort** *Am J Hum Genet* **109**:12–23 <https://doi.org/10.1016/j.ajhg.2021.11.008>
24. Choquet H, Paylakhi S, Kneeland SC, et al. 2018) **A multiethnic genome-wide association study of primary open-angle glaucoma identifies novel risk loci** *Nat Commun* **9**:2278 <https://doi.org/10.1038/s41467-018-04555-4>
25. Gao XR, Huang H, Nannini DR, Fan F, Kim H 2018) **Genome-wide association analyses identify new loci influencing intraocular pressure** *Hum Mol Genet* **27**:2205–2213 <https://doi.org/10.1093/hmg/ddy111>
26. Gharahkhani P, Jorgenson E, Hysi P, et al. 2021) **Genome-wide meta-analysis identifies 127 open-angle glaucoma loci with consistent effect across ancestries** *Nat Commun* **12**:1258 <https://doi.org/10.1038/s41467-020-20851-4>

27. Khawaja AP, Cooke Bailey JN, Wareham NJ, et al. 2018) **Genome-wide analyses identify 68 new loci associated with intraocular pressure and improve risk prediction for primary open-angle glaucoma** *Nat Genet* **50**:778–782 <https://doi.org/10.1038/s41588-018-0126-8>
28. MacGregor S, Ong JS, An J, et al. 2018) **Genome-wide association study of intraocular pressure uncovers new pathways to glaucoma** *Nat Genet* **50**:1067–1071 <https://doi.org/10.1038/s41588-018-0176-y>
29. Shiga Y, Akiyama M, Nishiguchi KM, et al. 2018) **Genome-wide association study identifies seven novel susceptibility loci for primary open-angle glaucoma** *Hum Mol Genet* **27**:1486–1496 <https://doi.org/10.1093/hmg/ddy053>
30. Cross SH, Macalinao DG, McKie L, et al. 2014) **A Dominant-Negative Mutation of Mouse Lmx1b Causes Glaucoma and Is Semi-lethal via LBD1-Mediated Dimerisation** *PLoS Genet* **10**:e1004359 <https://doi.org/10.1371/journal.pgen.1004359>
31. Tolman NG, Balasubramanian R, Macalinao DG, et al. 2021) **Genetic background modifies vulnerability to glaucoma-related phenotypes in Lmx1b mutant mice** *Dis Model Mech* **14**:dmm046953 <https://doi.org/10.1242/dmm.046953>
32. Mabuchi F, Mabuchi N, Sakurada Y, et al. 2017) **Additive effects of genetic variants associated with intraocular pressure in primary open-angle glaucoma** *PLOS One* **12**:e0183709 <https://doi.org/10.1371/journal.pone.0183709>
33. Tham YC, Liao J, Vithana EN, et al. 2015) **Aggregate Effects of Intraocular Pressure and Cup-to-Disc Ratio Genetic Variants on Glaucoma in a Multiethnic Asian Population** *Ophthalmology* **122**:1149–1157 <https://doi.org/10.1016/j.ophtha.2015.01.024>
34. Zanon-Moreno V, Ortega-Azorin C, Asensio-Marquez EM, et al. 2017) **A Multi-Locus Genetic Risk Score for Primary Open-Angle Glaucoma (POAG) Variants Is Associated with POAG Risk in a Mediterranean Population: Inverse Correlations with Plasma Vitamin C and E Concentrations** *Int J Mol Sci* **18**:2302 <https://doi.org/10.3390/ijms18112302>
35. Han X, Souzeau E, Ong JS, et al. 2019) **Myocilin Gene Gln368Ter Variant Penetrance and Association With Glaucoma in Population-Based and Registry-Based Studies** *JAMA Ophthalmol* **137**:28–35 <https://doi.org/10.1001/jamaophthalmol.2018.4477>
36. Gao XR, Huang H, Kim H 2019) **Polygenic Risk Score Is Associated With Intraocular Pressure and Improves Glaucoma Prediction in the UK Biobank Cohort** *Transl Vis Sci Technol* **8**:10 <https://doi.org/10.1167/tvst.8.2.10>
37. Nannini DR, Kim H, Fan F, Gao X 2018) **Genetic risk score is associated with vertical cup-disc ratio and improves prediction of primary open angle glaucoma in Latinos** *Ophthalmology* **125**:815–821 <https://doi.org/10.1016/j.ophtha.2017.12.014>
38. Zeleznik OA, Kang JH, Lasky-Su J, et al. 2023) **Plasma metabolite profile for primary open-angle glaucoma in three US cohorts and the UK Biobank** *Nat Commun* **14**:2860 <https://doi.org/10.1038/s41467-023-38466-w>
39. Madjedi KM, Stuart KV, Chua SY, et al. 2022) **The Association between Serum Lipids and Intraocular Pressure in 2 Large United Kingdom Cohorts** *Ophthalmology* **129**:986 <https://doi.org/10.1016/j.ophtha.2022.04.023>

40. Wang S, Hyperlipidemia Bao X. 2019) **Blood Lipid Level, and the Risk of Glaucoma: A Meta-Analysis** *Invest Ophthalmol Vis Sci* **60**:1028–1043 <https://doi.org/10.1167/iov.18-25845>
41. Pertl L, Mossböck G, Wedrich A, et al. 2017) **Triglycerides and Open Angle Glaucoma - A Meta-analysis with meta-regression** *Sci Rep* **7**:7829 <https://doi.org/10.1038/s41598-017-08295-1>
42. Rasoulinejad SA, Kasiri A, Montazeri M, et al. 2015) **The Association Between Primary Open Angle Glaucoma and Clustered Components of Metabolic Syndrome** *Open Ophthalmol J* **9**:149–155 <https://doi.org/10.2174/1874364101509010149>
43. Williams PA, Casson RJ 2024) **Glycolysis and glucose metabolism as a target for bioenergetic and neuronal protection in glaucoma** *Neural Regen Res* **19**:1637 <https://doi.org/10.4103/1673-5374.389638>
44. Harun-Or-Rashid M, Pappenhagen N, Zubricky R, Coughlin L, Jassim AH, Inman DM 2020) **MCT2 Overexpression Rescues Metabolic Vulnerability and Protects Retinal Ganglion Cells in Two Models of Glaucoma** *Neurobiol Dis* **141** <https://doi.org/10.1016/j.nbd.2020.104944>
45. Pappenhagen N, Harun-Or-Rashid M, Jassim Jaboori A, Inman DM 2019) **Upregulation of monocarboxylate transporter 2 protects retinal ganglion cells in glaucoma** *Invest Ophthalmol Vis Sci* **60**:622
46. Vohra R, Aldana BI, Bulli G, et al. 2019) **Lactate-Mediated Protection of Retinal Ganglion Cells** *J Mol Biol* **431**:1878–1888 <https://doi.org/10.1016/j.jmb.2019.03.005>
47. Rajala RVS, Rajala A 2023) **Unlocking the role of lactate: metabolic pathways, signaling, and gene regulation in postmitotic retinal cells** *Front Ophthalmol* **3** <https://doi.org/10.3389/fopht.2023.1296624>
48. Williams PA, Harder JM, John SWM 2017) **Glaucoma as a metabolic optic neuropathy: making the case for nicotinamide treatment in glaucoma** *J Glaucoma* **26**:1161–1168 <https://doi.org/10.1097/IJG.0000000000000767>
49. Inman DM, Harun-Or-Rashid M 2017) **Metabolic Vulnerability in the Neurodegenerative Disease Glaucoma** *Front Neurosci* **11** <https://doi.org/10.3389/fnins.2017.00146>
50. Williams PA, Harder JM, Foxworth NE, et al. 2017) **Vitamin B3 modulates mitochondrial vulnerability and prevents glaucoma in aged mice** *Science* **355**:756–760 <https://doi.org/10.1126/science.aal0092>
51. Chen S da, Wang L, Zhang X lan 2013) **Neuroprotection in glaucoma: present and future** *Chin Med J (Engl)* **126**:1567–1577
52. Vohra R, Dalgaard LM, Vibæk J, et al. 2019) **Potential metabolic markers in glaucoma and their regulation in response to hypoxia** *Acta Ophthalmol (Copenh)* **97**:567–576 <https://doi.org/10.1111/aos.14021>
53. Brooks GA 2020) **Lactate as a fulcrum of metabolism** *Redox Biol* **35**:101454 <https://doi.org/10.1016/j.redox.2020.101454>
54. Arnold PK, Finley LWS 2022) **Regulation and function of the mammalian tricarboxylic acid cycle** *J Biol Chem* **299**:102838 <https://doi.org/10.1016/j.jbc.2022.102838>

55. Jiménez-Moreno N, Kollareddy M, Stathakos P, et al. 2023) **ATG8-dependent LMX1B-autophagy crosstalk shapes human midbrain dopaminergic neuronal resilience** *J Cell Biol* **222**:e201910133 <https://doi.org/10.1083/jcb.201910133>
56. Kim SW, Lee HK, Shin JH, Lee JK 2013) **Up-down Regulation of HO-1 and iNOS Gene Expressions by Ethyl Pyruvate via Recruiting p300 to Nrf2 and Depriving It from p65** *Free Radic Biol Med* **65**:468–476 <https://doi.org/10.1016/j.freeradbiomed.2013.07.028>
57. Shin JH, Kim SW, Jin Y, Kim ID, Lee JK 2012) **Ethyl pyruvate-mediated Nrf2 activation and hemoxygenase 1 induction in astrocytes confer protective effects via autocrine and paracrine mechanisms** *Neurochem Int* **61**:89–99 <https://doi.org/10.1016/j.neuint.2012.04.005>
58. Harder JM, Guymer C, Wood JPM, et al. 2020) **Disturbed glucose and pyruvate metabolism in glaucoma with neuroprotection by pyruvate or rapamycin** *Proc Natl Acad Sci* **117**:33619–33627 <https://doi.org/10.1073/pnas.2014213117>
59. Guymer C, Chidlow G, Wood JPM, Casson RJ 2018) **Oral pyruvate supplementation protects against neurodegeneration in a rat model of glaucoma** *Invest Ophthalmol Vis Sci* **59**:2618
60. De Moraes CG, John SWM, Williams PA, Blumberg DM, Cioffi GA, Liebmann JM 2022) **Nicotinamide and Pyruvate for Neuroenhancement in Open-Angle Glaucoma** *JAMA Ophthalmol* **140**:11–18 <https://doi.org/10.1001/jamaophthalmol.2021.4576>
61. Wu X, Dai H, Liu L, et al. 2019) **Citrate reduced oxidative damage in stem cells by regulating cellular redox signaling pathways and represent a potential treatment for oxidative stress-induced diseases** *Redox Biol* **21** <https://doi.org/10.1016/j.redox.2018.11.015>
62. Morganti C, Bonora M, Marchi S, et al. 2020) **Citrate Mediates Crosstalk between Mitochondria and the Nucleus to Promote Human Mesenchymal Stem Cell In Vitro Osteogenesis** *Cells* **9**:1034 <https://doi.org/10.3390/cells9041034>
63. Catalina-Rodriguez O, Kolukula VK, Tomita Y, et al. 2012) **The mitochondrial citrate transporter, CIC, is essential for mitochondrial homeostasis** *Oncotarget* **3**:1220–1235 <https://doi.org/10.18632/oncotarget.714>
64. Ohanele C, Peoples JN, Karlstaedt A, et al. 2024) **The mitochondrial citrate carrier SLC25A1 regulates metabolic reprogramming and morphogenesis in the developing heart** *Commun Biol* **7**:1–16 <https://doi.org/10.1038/s42003-024-07110-8>
65. Chhimpa N, Singh N, Puri N, Kayath HP 2023) **The Novel Role of Mitochondrial Citrate Synthase and Citrate in the Pathophysiology of Alzheimer's Disease** *J Alzheimers Dis* **94**:S453–S472 <https://doi.org/10.3233/JAD-220514>
66. Fraenkl SA, Muser J, Groell R, et al. 2011) **Plasma Citrate Levels as a Potential Biomarker for Glaucoma** *J Ocul Pharmacol Ther* **27**:577–580 <https://doi.org/10.1089/jop.2011.0062>
67. Michalczyk M, Tadeusz P, Urban B, Anna W, Łazarczyk AB 2017) **Plasma citrate concentration: a possible biomarker for glaucoma in children** *BMJ Paediatr Open* **1**:e000023 <https://doi.org/10.1136/bmjpo-2017-000023>
68. Nusinovici S, Li H, Thakur S, et al. 2022) **High-Density Lipoprotein 3 Cholesterol and Primary Open-Angle Glaucoma: Metabolomics and Mendelian Randomization Analyses** *Ophthalmology* **129**:285–294 <https://doi.org/10.1016/j.ophtha.2021.09.013>

69. Masson EAY, Serrano J, Leger-Charnay E, Acar N (2024) **Cholesterol and oxysterols in retinal neuron-glia interactions: relevance for glaucoma** *Front Ophthalmol* **3**:1303649 <https://doi.org/10.3389/fopht.2023.1303649>
70. Colijn JM, den Hollander AI, Demirkan A, et al. **Increased High-Density Lipoprotein Levels Associated with Age-Related Macular Degeneration: Evidence from the EYE-RISK and European Eye Epidemiology Consortia** *Ophthalmology* **126**:393–406 <https://doi.org/10.1016/j.ophtha.2018.09.045>
71. Park SJ, Park DH (2020) **REvisiting Lipids in REtinal Diseases: A Focused Review on Age-related Macular Degeneration and Diabetic Retinopathy** *J Lipid Atheroscler* **9**:406 <https://doi.org/10.12997/jla.2020.9.3.406>
72. Pasquale LR, Khawaja AP, Wiggs JL, et al. (2023) **Metabolite and Lipid Biomarkers Associated With Intraocular Pressure and Inner Retinal Morphology: 1 H NMR Spectroscopy Results From the UK Biobank** *Investig Ophthalmology Vis Sci* **64**:11 <https://doi.org/10.1167/iovs.64.11.11>
73. Ana R da, Gliszczyńska A, Sanchez-Lopez E, et al. (2023) **Precision Medicines for Retinal Lipid Metabolism-Related Pathologies** *J Pers Med* **13**:635 <https://doi.org/10.3390/jpm13040635>
74. Fu Z, Chen CT, Cagnone G, et al. (2019) **Dyslipidemia in retinal metabolic disorders** *EMBO Mol Med* **11**:e10473 <https://doi.org/10.15252/emmm.201910473>
75. Yuksel S, Ufret-Vincenty R, Butovich IA (2023) **The role of Lipids in lipid metabolism in the retina** *Invest Ophthalmol Vis Sci* **64**:5454
76. Kelly UL, Grigsby D, Cady MA, et al. (2020) **High-density lipoproteins are a potential therapeutic target for age-related macular degeneration** *J Biol Chem* **295**:13601–13616 <https://doi.org/10.1074/jbc.RA119.012305>
77. Morvaridzadeh M, Zoubdane N, Heshmati J, Alami M, Berrougui H, Khalil A (2024) **High-Density Lipoprotein Metabolism and Function in Cardiovascular Diseases: What about Aging and Diet Effects?** *Nutrients* **16**:653 <https://doi.org/10.3390/nu16050653>
78. Hamel AR, Yan W, Rouhana JM, et al. (2024) **Integrating genetic regulation and single-cell expression with GWAS prioritizes causal genes and cell types for glaucoma** *Nat Commun* **15**:396 <https://doi.org/10.1038/s41467-023-44380-y>
79. Minassian DC, Reidy A, Coffey M, Minassian A (2000) **Utility of predictive equations for estimating the prevalence and incidence of primary open angle glaucoma in the UK** *Br J Ophthalmol* **84**:1159–1161 <https://doi.org/10.1136/bjo.84.10.1159>
80. Sudlow C, Gallacher J, Allen N, et al. (2015) **UK Biobank: An Open Access Resource for Identifying the Causes of a Wide Range of Complex Diseases of Middle and Old Age** *PLoS Med* **12**:e1001779 <https://doi.org/10.1371/journal.pmed.1001779>
81. Bycroft C, Freeman C, Petkova D, et al. (2018) **The UK Biobank resource with deep phenotyping and genomic data** *Nature* **562**:203–209 <https://doi.org/10.1038/s41586-018-0579-z>
82. Zeleznik O, Kang JH, Lasky-Su J, et al. (2022) **Plasma metabolomics of primary open-angle glaucoma in three prospective US cohorts and the UK Biobank** *Published online February 28* <https://doi.org/10.1101/2022.02.24.22271483>

83. Elliott P (2008) **Peakman TC, on behalf of UK Biobank. The UK Biobank sample handling and storage protocol for the collection, processing and archiving of human blood and urine** *Int J Epidemiol* **37**:234–244 <https://doi.org/10.1093/ije/dym276>
84. Julkunen H, Cichońska A, Slagboom PE, Würtz P, Nightingale Health UK Biobank Initiative (2021) **Metabolic biomarker profiling for identification of susceptibility to severe pneumonia and COVID-19 in the general population** *eLife* **10**:e63033 <https://doi.org/10.7554/eLife.63033>
85. Takis PG, Ghini V, Tenori L, Turano P, Luchinat C (2019) **Uniqueness of the NMR approach to metabolomics** *TrAC Trends Anal Chem* **120**:115300 <https://doi.org/10.1016/j.trac.2018.10.036>
86. Soininen P, Kangas AJ, Würtz P, Suna T, Ala-Korpela M (2015) **Quantitative serum nuclear magnetic resonance metabolomics in cardiovascular epidemiology and genetics** *Circ Cardiovasc Genet* **8**:192–206 <https://doi.org/10.1161/CIRCGENETICS.114.000216>
87. Mihaleva VV, Korhonen SP, van Duynhoven J, Niemitz M, Vervoort J, Jacobs DM (2014) **Automated quantum mechanical total line shape fitting model for quantitative NMR-based profiling of human serum metabolites** *Anal Bioanal Chem* **406**:3091–3102 <https://doi.org/10.1007/s00216-014-7752-5>
88. Kim J, Aschard H, Kang JH, et al. (2020) **and Dietary Caffeine Consumption: A Gene—Diet Interaction Study from the UK Biobank** *Ophthalmology* **128**:866 <https://doi.org/10.1016/j.ophttha.2020.12.009>
89. Gao X, Starmer J, Martin ER (2008) **A multiple testing correction method for genetic association studies using correlated single nucleotide polymorphisms** *Genet Epidemiol* **32**:361–369 <https://doi.org/10.1002/gepi.20310>
90. Thaung C, West K, Clark BJ, et al. (2002) **Novel ENU-induced eye mutations in the mouse: models for human eye disease** *Hum Mol Genet* **11**:755–767 <https://doi.org/10.1093/hmg/11.7.755>
91. John SW, Hagaman JR, MacTaggart TE, Peng L, Smithes O (1997) **Intraocular pressure in inbred mouse strains** *Invest Ophthalmol Vis Sci* **38**:249–253
92. Savinova OV, Sugiyama F, Martin JE, et al. (2001) **Intraocular pressure in genetically distinct mice: an update and strain survey** *BMC Genet* **2** <https://doi.org/10.1186/1471-2156-2-12>
93. Howell GR, Libby RT, Jakobs TC, et al. (2007) **Axons of retinal ganglion cells are insulted in the optic nerve early in DBA/2J glaucoma** *J Cell Biol* **179**:1523–1537 <https://doi.org/10.1083/jcb.200706181>
94. Nair KS, Cosma M, Raghupathy N, et al. (2016) **YBR/Eij mice: a new model of glaucoma caused by genes on chromosomes 4 and 17** *Dis Model Mech* **9**:863–871 <https://doi.org/10.1242/dmm.024307>
95. Howell GR, Soto I, Zhu X, et al. (2012) **Radiation treatment inhibits monocyte entry into the optic nerve head and prevents neuronal damage in a mouse model of glaucoma** *J Clin Invest* **122**:1246–1261 <https://doi.org/10.1172/JCI61135>

96. Howell GR, Macalinao DG, Sousa GL, et al. 2011) **Molecular clustering identifies complement and endothelin induction as early events in a mouse model of glaucoma** *J Clin Invest* **121**:1429–1444 <https://doi.org/10.1172/JCI44646>

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Reviewer #1 (Public review):

Summary:

The Authors explore associations between plasma metabolites and glaucoma, a primary cause of irreversible vision loss worldwide. The study relies on measurements of 168 plasma metabolites in 4,658 glaucoma patients and 113,040 controls from the UK Biobank. The Authors show that metabolites improve the prediction of glaucoma risk based on polygenic risk score (PRS) alone, albeit weakly. The Authors also report a "metabolomic signature" that is associated with a reduced risk (or "resilience") for developing glaucoma among individuals in the highest PRS decile (reduction of risk by an estimated 29%). The Authors highlight the protective effect of pyruvate, a product of glycolysis, for glaucoma development and show that this molecule mitigates elevated intraocular pressure and optic nerve damage in a mouse model of this disease.

Strengths:

This work provides additional evidence that glycolysis may play a role in the pathophysiology of glaucoma. Previous studies have demonstrated the existence of an inverse relationship between intraocular pressure and retinal pyruvate levels in animal models (Hader et al. 2020, PNAS 117(52)) and pyruvate supplementation is currently being explored for neuro-enhancement in patients with glaucoma (De Moraes et al. 2022, JAMA Ophthalmology 140(1)). The study design is rigorous and relies on validated standard methods. Additional insights gained from a mouse model are valuable.

Weaknesses:

Caution is warranted when examining and interpreting the results of this study. Among all participants (cases and controls) glaucoma status was self-reported, determined on the basis of ICD codes or previous glaucoma laser/surgical therapy. This is problematic as it is not uncommon for individuals in the highest PRS decile to have undiagnosed glaucoma (as shown in previous work by some of the authors of this article). The Authors acknowledge a

"relatively low glaucoma prevalence in the highest decile group" but do not explore how undiagnosed glaucoma may affect their results. This also applies to all controls selected for this study. The Authors state that "50 to 70% of people affected [with glaucoma] remain undiagnosed". Therefore, the absence of self-reported glaucoma does not necessarily indicate that the disease is not present. Validation of the findings from this study in humans is, therefore, critical. This should ideally be performed in a well-characterized glaucoma cohort, in which case and control status has been assessed by qualified clinicians.

The authors indicate that within the top decile of PRS participants with glaucoma are more likely to be of white ethnicity, while they are more likely to be of Black and Asian ethnicity if they are in the bottom half of PRS. Have the Authors explored how sensitive their predictions are to ethnicity? Since their cohort is predominantly of European ancestry (85.8%), would it make sense to exclude other ethnicities to increase the homogeneity of the cohort and reduce the risk for confounders that may not be explicitly accounted for?

The authors discuss the importance of pyruvate, and lactate for retinal ganglion cell survival along with that of several lipoproteins for neuroprotection. However, there is a distinction to be made between locally produced/available glycolysis end products and lipoproteins and those circulating in the blood. It may be useful to discuss this in the manuscript, and for the Authors to explore if plasma metabolites may be linked to metabolism that takes place past the blood-retinal barrier.

Comments on revisions:

The Authors have addressed all of my concerns.

<https://doi.org/10.7554/eLife.105576.2.sa2>

Reviewer #2 (Public review):

Summary

The authors have used the UK Biobank data to interrogate the association between plasma metabolites and glaucoma.

- (1) They initially assessed plasma metabolites as predictors of glaucoma: The addition of NMR-derived metabolomic data to existing models containing clinical and genetic data was marginal.
- (2) They then determined whether certain metabolites might protect against glaucoma in individuals at high genetic risk: Certain molecules in bioenergetic pathways (lactate, pyruvate and citrate) conferred protection.
- (3) They provide support for protection conferred by pyruvate in a murine model.

Weaknesses

- (1) Although it is an invaluable treasure trove of data, selection bias and self-reporting are inescapable problems when using the UK Biobank data for glaucoma research. The high-impact glaucoma-related GWAS publications (Ref 26 and 27) referenced in support of the method suffer the same limitations. This doesn't negate the conclusions but must be taken into consideration. The authors might note that it is somewhat reassuring that the proportion of glaucoma cases (4%) is close to what would be expected in a population-based study of 40-69-year-olds of predominantly white ethnicity.
- (2) As noted by the authors, a limitation is the predominantly white ethnicity profile that comprises the UK Biobank.
- (3) Also as noted by the authors, the study is cross-sectional and is limited by the "correlation does not imply causation" issue.

(4) The optimal collection, transport and processing of the samples for NMR metabolite analysis is critical for accurate results. Strict policies were in place for these procedures, but deviations from protocol remain an unknown influence on the data.

(5) In addition, all UK Biobank blood samples had unintended dilution during the initial sample storage process at UK Biobank facilities. (Julkunen, H. et al. Atlas of plasma NMR biomarkers for health and disease in 118,461 individuals from the UK Biobank. Nat Commun 14, 604 (2023) Samples from aliquot 3, used for the NMR measurements, suffered from 5-10% dilution. (Allen, Naomi E., et al. Wellcome Open Research 5 (2021): 222.) Julkunen et al. report that "The dilution is believed to come from mixing of participant samples with water due to seals that failed to hold a system vacuum in the automated liquid handling systems. While this issue is likely to have an impact on some of the absolute biomarker concentration values, it is expected to have limited impact on most epidemiological analyses."

Strengths

The huge sample size supports a powerful statistical analysis and the opportunity for the inclusion of multiple covariates and interactions without overfitting the models.

The authors have constructed a robust methodology and statistical design.

The manuscript is well-written, and the study is logically presented.

The Figures are of good quality.

Broadly, the conclusions are justified by the findings.

Impact

The findings advance personalized prognostics for glaucoma that combine metabolomic and genetic data. In addition, the protective effect of certain metabolites influences further research on novel therapeutic strategies.

Comments on revisions:

The authors have thoughtfully and comprehensively addressed my comments. I have no further comments.

<https://doi.org/10.7554/eLife.105576.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

The authors explore associations between plasma metabolites and glaucoma, a primary cause of irreversible vision loss worldwide. The study relies on measurements of 168 plasma metabolites in 4,658 glaucoma patients and 113,040 controls from the UK Biobank. The authors show that metabolites improve the prediction of glaucoma risk based on polygenic risk score (PRS) alone, albeit weakly. The authors also report a "metabolomic signature" that is associated with a reduced risk (or "resilience") for developing glaucoma among individuals in the highest PRS decile (reduction of risk by an estimated 29%). The authors highlight the protective effect of pyruvate, a product of glycolysis, for glaucoma development and show that this molecule mitigates elevated intraocular pressure and optic nerve damage in a mouse model of this disease.

Strengths:

This work provides additional evidence that glycolysis may play a role in the pathophysiology of glaucoma. Previous studies have demonstrated the existence of an inverse relationship between intraocular pressure and retinal pyruvate levels in animal models (Hader et al. 2020, PNAS 117(52)) and pyruvate supplementation is currently being explored for neuro-enhancement in patients with glaucoma (De Moraes et al. 2022, JAMA Ophthalmology 140(1)). The study design is rigorous and relies on validated, standard methods. Additional insights gained from a mouse model are valuable.

We thank the reviewer for these supportive comments.

Weaknesses:

Caution is warranted when examining and interpreting the results of this study. Among all participants (cases and controls) glaucoma status was self-reported, determined on the basis of ICD codes or previous glaucoma laser/surgical therapy. This is problematic as it is not uncommon for individuals in the highest PRS decile to have undiagnosed glaucoma (as shown in previous work by some of the authors of this article). The authors acknowledge a "relatively low glaucoma prevalence in the highest decile group" but do not explore how undiagnosed glaucoma may affect their results. This also applies to all controls selected for this study. The authors state that "50 to 70% of people affected [with glaucoma] remain undiagnosed". Therefore, the absence of self-reported glaucoma does not necessarily indicate that the disease is not present. Validation of the findings from this study in humans is, therefore, critical. This should ideally be performed in a well-characterized glaucoma cohort, in which case and control status has been assessed by qualified clinicians.

We appreciate the comment regarding the challenges of glaucoma ascertainment in UK Biobank. This is a valid limitation, as glaucoma in UK Biobank is based on self-reports and hospital records rather than comprehensive ophthalmologic examinations for all participants. To the best of our knowledge, there is no comparably sized dataset where all participants have undergone standardized glaucoma assessments, comprehensive metabolomic profiling, and high-throughput genotyping. Work is currently ongoing to link UK Biobank data to ophthalmic EMR data, which will help confirm self-reported diagnoses. This work is not complete, and the coverage of the cohort from such linkage is uncertain at present. Nonetheless, several factors speak to the validity of our findings. The top members of the metabolomic signature associated with resilience in the top decile of glaucoma polygenic risk score (PRS) decile—lactate ($P=8.8E-12$) and pyruvate ($P=1.9E-10$)—had robust values for statistical significance after appropriate adjustment for multiple comparisons, with additional validation for pyruvate in a human-relevant, glaucoma mouse model. Strikingly, the glaucoma odds ratio (OR) for subjects in the highest quartile of glaucoma PRS and metabolic risk score (MRS) was 25-fold, using participants in the lowest quartile of glaucoma PRS and MRS as the reference group. An effect size this large for a putative glaucoma determinant has only been seen for intraocular pressure (IOP), which is now largely accepted to be in the causal pathway of the disease.

The Discussion now contains the following statement: "A second limitation is that glaucoma ascertainment in the UK Biobank is based on self-reported diagnoses and hospital records rather than comprehensive ophthalmologic examinations. Nonetheless, it is reassuring that the prevalence of glaucoma in our sample (~4%) is similar to a directly performed disease burden estimate in a comparable, albeit slightly older, United Kingdom sample (2.7%)(79)". (Lines 379-382)

The authors indicate that within the top decile of PRS participants with glaucoma are more likely to be of white ethnicity, while they are more likely to be of Black and Asian

ethnicity if they are in the bottom half of PRS. Have the authors explored how sensitive their predictions are to ethnicity? Since their cohort is predominantly of European ancestry (85.8%), would it make sense to exclude other ethnicities to increase the homogeneity of the cohort and reduce the risk for confounders that may not be explicitly accounted for?

Comparing data in Tables 3 and 4 of the manuscript, we observe that, on a percentage basis, more individuals have glaucoma in the highest 10th percentile of risk compared to the lowest 50th percentile of risk across all ancestral groups. We recently reported that the risk of glaucoma increases with each standard deviation increase in the glaucoma PRS across ancestral groups in the UK Biobank, utilizing a slightly different sample size (see Author response table 1 below). (1) Since the PRS is applicable across ancestral groups, we aim to make our results as generalizable as possible; therefore, we prefer to report our findings for all ethnic groups and not restrict our results to Europeans.

Author response table 1.

Performance of the mtGPRS Across Ancestral Groups in the UK Biobank

Ancestry	N, glaucoma cases	N, controls	N, total	OR [95% CI]
African	184	2,448	2,632	1.25 [0.97 – 1.60]
Asian	199	4,475	4,674	1.63 [1.34 – 1.98]
UK European	7,973	162,190	170,163	2.84 [2.73 – 2.92]
Other European	220	5,837	6,057	1.67 [1.43 – 1.96]

Abbreviations: mtGPRS, multitrait analysis of GWAS polygenic risk score; OR, odds ratio; CI, confidence interval.(1)

UK Biobank ancestry was genetically inferred based on principal component analysis. The OR represents the risk associated with each standard deviation change in mtGRS and is adjusted for multiple covariates including age, sex, and medical comorbidities.

In the discussion, we stated that “... we chose to analyze Europeans and non-Europeans together to make the results as generalizable as possible.” (Lines 378-379)

The authors discuss the importance of pyruvate, and lactate for retinal ganglion cell survival, along with that of several lipoproteins for neuroprotection. However, there is a distinction to be made between locally produced/available glycolysis end products and lipoproteins and those circulating in the blood. It may be useful to discuss this in the manuscript, and for the authors to explore if plasma metabolites may be linked to metabolism that takes place past the blood-retinal barrier.

As the reviewer points out, it is crucial to interpret the results for lipoproteins within the context of their access to the blood-retinal barrier. Even for smaller metabolites like pyruvate and lactate, it is essential to consider local production versus serum-derived molecules in mediating any neuroprotective effects. Our murine data suggest that exogenous pyruvate contributed to neuroprotection. However, for the other glycolysis-related metabolites (lactate and citrate), we cannot rule out the possibility that locally produced metabolites may also contribute to neuroprotection. None of the lipoproteins identified as potential resilience biomarkers had an adjusted P-value of less than 0.05. Nevertheless, HDL analytes can cross

blood-ocular barriers to enter the aqueous humor.(2) Therefore, it is also possible for serum-derived HDL to influence retinal ganglion cell homeostasis. Overall, much more research is needed to clarify the roles of locally produced versus serum-derived factors in conferring resilience to genetic predisposition to glaucoma.

We have added the following sentences to the discussion:

“Notably, although our validation data confirm the neuroprotective effects of exogenous pyruvate, it remains possible that endogenously produced pyruvate within ocular tissues may also contribute to RGC protection.” (Lines 329-331)

“Furthermore, as HDL analytes can cross blood-ocular barriers,(78) there is a plausible route for serum-derived HDL to influence RGC homeostasis. Nonetheless, the relative contributions of circulating lipoproteins versus local synthesis within ocular tissues remain unclear and warrant further investigation.” (Lines 355-358)

“Incorporating ocular physiology and blood-retinal barrier considerations when interpreting lipoproteins as potential resilience biomarkers will be critical for future studies aimed at understanding and therapeutically targeting increased glaucoma risk.” (Lines 360-363)

Reviewer #2 (Public review):

Summary

The authors have used the UK Biobank data to interrogate the association between plasma metabolites and glaucoma.

(1) They initially assessed plasma metabolites as predictors of glaucoma: The addition of NMR-derived metabolomic data to existing models containing clinical and genetic data was marginal.

(2) They then determined whether certain metabolites might protect against glaucoma in individuals at high genetic risk: Certain molecules in bioenergetic pathways (lactate, pyruvate, and citrate) conferred protection.

(3) They provide support for protection conferred by pyruvate in a murine model.

Strengths

(1) The huge sample size supports a powerful statistical analysis and the opportunity for the inclusion of multiple covariates and interactions without overfitting the models.

(2) The authors have constructed a robust methodology and statistical design.

(3) The manuscript is well written, and the study is logically presented.

(4) The figures are of good quality.

(5) Broadly, the conclusions are justified by the findings.

We thank the reviewer for these supportive comments.

Weaknesses

(1) Although it is an invaluable treasure trove of data, selection bias and self-reporting are inescapable problems when using the UK Biobank data for glaucoma research. The high-impact glaucoma-related GWAS publications (references 26 and 27) referenced in support of the method suffer the same limitations. This doesn't negate the conclusions but must be taken into consideration. The authors might note that it is somewhat

reassuring that the proportion of glaucoma cases (4%) is close to what would be expected in a population-based study of 40-69-year-olds of predominantly white ethnicity.

While there are limitations when open-angle glaucoma (OAG) is ascertained by self-report, as discussed above, we agree with the reviewer that the prevalence of glaucoma is consistent with data from population-based studies of Europeans who are 40-69 years of age.

We also want to point out that references 26 and 27 indicate glaucoma self-reports can be an acceptable surrogate for OAG that is ascertained by clinical evaluation. Consider the methodologic details for each study:

Reference 26 is a 4-stage genome-wide meta-analysis to identify loci for OAG from 21 independent populations. The phenotypic definition of OAG was based on clinical assessment in the discovery stage, and 7286 glaucoma self-reports from the UK Biobank served as an effective replication set. It is also important to note that 120 out of the 127 discovered OAG loci were nominally replicated in 23andMe, where glaucoma was ascertained entirely by self-report.

Reference 27 is a genome-wide meta-analysis to identify IOP genetic loci, an important endophenotype for OAG. The study identified 112 loci for IOP. These loci were incorporated into a glaucoma prediction model in the NEIGHBORHOOD study and the UK Biobank. The area under the receiver operator curve was 0.76 and 0.74, respectively, in these studies. While the AUCs were similar, OAG was ascertained clinically in NEIGHBORHOOD and largely by self-report in UK Biobank.

Finally, a strength of the UK Biobank is that selection bias is minimized. Patients need not be insured or aligned to the study for any reason aside from being a UK resident. There is indeed a healthy bias in the UK Biobank. Ambulatory patients who tend to be health conscious and willing to donate their time and provide biological specimens tend to participate. We agree with the reviewer that the use of self-reported cases does not negate the conclusions, and hopefully, future iterations of the UK Biobank where clinical validation of self-reports are performed will confirm these findings, which already have some validation in a preclinical glaucoma model.

We add the following sentence to the first action item above regarding our case ascertainment method. “Nonetheless, it is reassuring that the prevalence of glaucoma in our sample (~4%) is similar to a directly performed disease burden estimate in a comparable, albeit slightly older, United Kingdom sample (2.7%)..”(3) (Lines 381-383)

(2) As noted by the authors, a limitation is the predominantly white ethnicity profile that comprises the UK Biobank.

(3) Also as noted by the authors, the study is cross-sectional and is limited by the "correlation does not imply causation" issue.

While the epidemiological arm of our study was cross-sectional, the studies testing the ability of pyruvate to mitigate the glaucoma phenotype in mice with the *Lmx1b* mutation were prospective.

We already pointed out in the discussion that pyruvate supplementation reduced glaucoma incidence in a human-relevant genetic mouse model.

(4) The optimal collection, transport, and processing of the samples for NMR metabolite analysis is critical for accurate results. Strict policies were in place for these procedures, but deviations from protocol remain an unknown influence on the data.

Comments 4 and 5 are related and will be addressed after comment 5.

(5) In addition, all UK Biobank blood samples had unintended dilution during the initial sample storage process at UK Biobank facilities. (Julkunen, H. et al. Atlas of plasma NMR biomarkers for health and disease in 118,461 individuals from the UK Biobank. Nat Commun 14, 604 (2023) Samples from aliquot 3, used for the NMR measurements, suffered from 5-10% dilution. (Allen, Naomi E., et al. Wellcome Open Research 5 (2021): 222.) Julkunen et al. report that "The dilution is believed to come from mixing of participant samples with water due to seals that failed to hold a system vacuum in the automated liquid handling systems. While this issue is likely to have an impact on some of the absolute biomarker concentration values, it is expected to have limited impact on most epidemiological analyses."

We thank the reviewer for making us aware of the unintended sample dilution issue from aliquot 3, used for NMR metabolomics in UK Biobank participants. While ~98% of samples experienced a 5-10% dilution, this would not affect our reported results, which did not rely on absolute biomarker concentrations. All metabolites in the main tables were probit transformed and used as continuous variables per 1 standard deviation increase. Nonetheless, in supplemental material, we show the unadjusted median levels of pyruvate (in mmol/L) were higher in participants without glaucoma vs those with glaucoma, both in the population overall and in those in the top 10 percentile of glaucoma risk.

Furthermore, we see no evidence in the literature that unidentified protocol deviations might impact metabolite results in UK Biobank participants. For example, a recent study evaluated the relationship between a weighted triglyceride-raising polygenic score (TG.PS) and type 3 hyperlipidemia (T3HL) in the Oxford Biobank (OBB) and the UK Biobank. In both biobanks, metabolomics was performed on the Nightingale NMR platform. A one standard deviation increase in TG.PS was associated with a 13% and 15.2% increased risk of T3HL in the OBB and UK Biobank, respectively.(4) Replication of the OBB result in the UK Biobank suggests there are no additional concerns regarding the processing of the UK Biobank for NMR metabolomics. Of course, we remain vigilant for protocol deviations that might call our results into question and will seek to validate our findings in other biobanks in future research.

Impact

The findings advance personalized prognostics for glaucoma that combine metabolomic and genetic data. In addition, the protective effect of certain metabolites influences further research on novel therapeutic strategies.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Given the uncertainty in the proportion of controls with undiagnosed glaucoma, it may be appropriate to include a sensitivity analysis in the manuscript. The authors could then provide the readers with an estimate of how sensitive their predictions are to the proportion of undiagnosed individuals among controls.

Since UK Biobank participants did not undergo standardized clinical assessments, it is not possible to perform sensitivity analyses as we don't know which controls might have glaucoma, although we can offer the following comments.

We are performing a cross-sectional, prospective, detailed glaucoma assessment of participants in the top and bottom 10 percent of genetic predisposition recruited from BioMe

at Icahn School of Medicine at Mount Sinai and Mass General Brigham Biobank at Harvard Medical School. We find that 21% of people in the top decile of genetic risk have glaucoma,(5) which compares reasonably well to the 15% of people in the top 10% of genetic risk in the UK Biobank. This underscores the assertion that our definition of glaucoma in the UK Biobank, while not ideal, is a reasonable surrogate for a detailed clinical assessment.

Currently, 10,077 subjects in the top decile of glaucoma genetic predisposition did not meet our definition of glaucoma. If we assume that the glaucoma prevalence is 3% and 50% of people with glaucoma are undiagnosed, then that would translate to an additional 150 cases misclassified as controls, which could either drive our result to the null, have no impact on our current result or contribute to a false positive result, depending on their pyruvate (and other metabolite) levels.

We have already addressed the issue of a lack of standardized exams in the UK Biobank and the need for more studies to confirm our findings.

Reviewer #2 (Recommendations for the authors):

(1) I am curious about the proposed reason for some individuals having metabolic profiles conferring resilience. Plasma pyruvate levels are normally distributed. Is it simply the case that some individuals with naturally high levels of pyruvate are fortuitously protected against glaucoma? Some sort of self-regulation mechanism seems unlikely.

Thank you for your insightful question regarding the potential mechanism underlying the association between pyruvate levels and glaucoma resilience. There may be modest inter-individual differences which can have significant physiological implications, particularly in the context of neurodegeneration and metabolic stress. One possibility is that individuals with naturally higher pyruvate levels may benefit from pyruvate's known neuroprotective and metabolic support functions(6–8), which could confer resilience against the oxidative and bioenergetic challenges associated with glaucoma. Pyruvate is important for cellular metabolism, redox balance, and mitochondrial function - processes that are increasingly implicated in glaucomatous neurodegeneration. (9)Elevated pyruvate levels support mitochondrial ATP production(10), buffer oxidative stress,(11) and impact metabolic flux(12,13) through pathways such as the tricarboxylic acid cycle and NAD⁺/NADH homeostasis. This is consistent with prior studies suggesting that mitochondrial dysfunction contributes to retinal ganglion cell vulnerability in glaucoma.(14–17) While a direct self-regulation mechanism may seem unlikely, both genetic and environmental factors can influence pyruvate metabolism, which could lead to subtle but clinically meaningful variations in its levels. Our findings are supported by validation in a mouse model, which suggests that the association is less likely fortuitous, but there may be an underlying biological process that merits further mechanistic investigation. Future studies incorporating longitudinal metabolic profiling and functional validation in human-derived models will help better understand this relationship.

(2) Conceivably, the higher levels of pyruvate and lactate may have resulted from recent exercise and may reflect individuals with high levels of exercise that confers resilience against glaucoma by independent mechanisms such as improved blood flow. Any way to rule that out from the UK Biobank data?

Thank you for raising this important point. To account for the potential confounding effects of physical activity, we adjusted for metabolic equivalents of task (METs) in our models, a standardized measure of physical activity available in the UK Biobank. By incorporating METs as a covariate, we aimed to minimize the influence of individual exercise levels on plasma pyruvate and lactate levels. This helps us ascertain that the observed associations are

not solely attributable to differences in physical activity. However, we do acknowledge that longitudinal analysis of exercise patterns would provide further clarity on this relationship.

(3) It may be worth mentioning that the retinal ganglion cells contain a plasma membrane monocarboxylate transporter that supports pyruvate and lactate uptake from the extracellular space.

Thank you for this extremely insightful suggestion on retinal ganglion cell (RGC) expression of monocarboxylate transporters, which can facilitate the uptake of pyruvate and lactate from the extracellular space. This is relevant for our study, given the high metabolic demands of RGCs and their reliance on both glycolytic and oxidative metabolism for neuroprotection and survival.

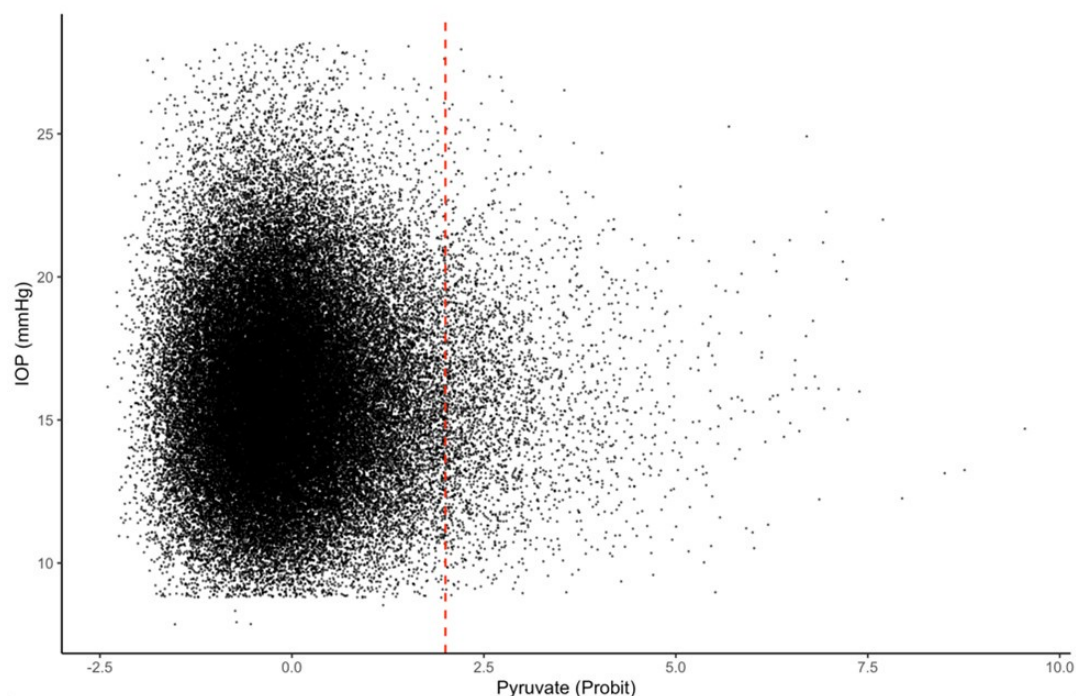
We acknowledged this in the discussion section of the manuscript by adding the following statement: "RGCs express monocarboxylate transporters, which facilitate the uptake of extracellular pyruvate and lactate, improving energy homeostasis, neuronal metabolism, and survival." (Lines 309-311)

(4) The mechanism of protection in the mice, at least in part, is likely due to the lower IOP in the pyruvate-treated animals. Did the authors investigate the influence of pyruvate on IOP in the UK Biobank data (about 110,000 individuals had IOP measurements)?

Thank you for your suggested investigation. We ran the suggested analysis among 68,761 individuals with IOP measurements and metabolomic profiling. Imputed pretreatment IOP values for participants using ocular hypotensive agents were calculated by dividing the measured IOP by 0.7, based on the mean IOP.

We plotted the relationship between IOP and pyruvate levels (probit transformed). We compared participants with pyruvate levels +2 standard deviations, above the mean (red dashed line), which has a probit-transformed value of 2 and an absolute concentration of 0.15 mmol/L. We found a statistically significant difference between the groups ($p=0.017$) using the Welch two-sample t-test. We have not added this analysis to the manuscript, but readers can find the data here as the reviews are public. We acknowledge and addressed the dilutional issue above, where we utilized probit-transformed metabolite levels analyzed as continuous variables per 1 SD increase, rather than absolute concentrations.

Author response image 1.



(5) Line 88: I suggest changing "patients" to "affected individuals". The term "patients" tends to imply that the individual has already been diagnosed, but the idea being conveyed is about underdiagnosis in the population.

Thank you for your suggestion.

We have added the change from "patients" to "affected individuals" in the introduction. (Line 90)

(6) Line 93: "However, glaucoma is also significantly affected by environmental and lifestyle factors, 10-14". Although lifestyle risk factors such as caffeine intake, alcohol, smoking, and air pollution have been reported, the associations are generally weak and inconsistently reported. Consider modifying this notion to stress the emerging evidence around gene-environment interactions (reference 14) rather than environmental factors per se, with the implication that genes + metabolism may be greater than the sum of the parts.

Thank you for this thoughtful suggestion to highlight gene-environment interactions, where genetic susceptibility may amplify or mitigate the impact of metabolic and environmental influences on glaucoma progression. We have revised the statement to better reflect the synergistic effects of genetics and metabolism rather than considering environmental factors in isolation.

Revised sentence for inclusion in the introduction of the manuscript: "Glaucoma risk is influenced by both genetic and metabolic factors, with emerging evidence suggesting that gene-environment interactions may play a greater role in conferring disease risk than independent exposures alone." (Lines 95-97)

(7) Lines 156-161: *In model 4, rather than stating that the very small increase in AUC with the addition of metabolic data compared to clinical and genetic data alone, "modestly enhances the prediction of glaucoma", it may be better interpreted as a marginal difference that was statistically significant due to the very large sample size but not clinically significant.*

Thank you for your suggested comment.

We have adjusted the wording by changing “modestly” to “marginally” to address that the statistical significance is in the context of the study’s large sample size in the results section (Line 162) and throughout the manuscript.

NB: We made other minor edits to correct minor grammatical errors, improve clarity, and streamline the revised manuscript. Furthermore, the paragraph regarding slit lamp examination in the Methods was inadvertently omitted but is added back in the revised manuscript (Lines 571-579).

References:

- (1) Kim J, Kang JH, Wiggs JL, et al. Does Age Modify the Relation Between Genetic Predisposition to Glaucoma and Various Glaucoma Traits in the UK Biobank? *Invest Ophthalmol Vis Sci.* 2025;66(2):57. doi:10.1167/iovs.66.2.57
- (2) Cenedella RJ. Lipoproteins and lipids in cow and human aqueous humor. *Biochim Biophys Acta BBA - Lipids Lipid Metab.* 1984;793(3):448-454. doi:10.1016/0005-2760(84)90262-5
- (3) Minassian DC, Reidy A, Coffey M, Minassian A. Utility of predictive equations for estimating the prevalence and incidence of primary open angle glaucoma in the UK. *Br J Ophthalmol.* 2000;84(10):1159-1161. doi:10.1136/bjo.84.10.1159
- (4) Pieri K, Trichia E, Neville MJ, et al. Polygenic risk in Type III hyperlipidaemia and risk of cardiovascular disease: An epidemiological study in UK Biobank and Oxford Biobank. *Int J Cardiol.* 2023;373:72-78. doi:10.1016/j.ijcard.2022.11.024
- (5) Zhao H, Pasquale LR, Zebardast N, et al. Screening by glaucoma polygenic risk score to identify primary open-angle glaucoma in two biobanks: An updated report. *ARVO 2025 meeting.* Published online 2025.
- (6) Zilberter Y, Gubkina O, Ivanov AI. A unique array of neuroprotective effects of pyruvate in neuropathology. *Front Neurosci.* 2015;9. doi:10.3389/fnins.2015.00017
- (7) Quansah E, Peelaerts W, Langston JW, Simon DK, Colca J, Brundin P. Targeting energy metabolism via the mitochondrial pyruvate carrier as a novel approach to attenuate neurodegeneration. *Mol Neurodegener.* 2018;13(1):28. doi:10.1186/s13024-018-0260-x
- (8) Gray LR, Tompkins SC, Taylor EB. Regulation of pyruvate metabolism and human disease. *Cell Mol Life Sci.* 2014;71(14):2577-2604. doi:10.1007/s00018-013-1539-2
- (9) Harder JM, Guymer C, Wood JPM, et al. Disturbed glucose and pyruvate metabolism in glaucoma with neuroprotection by pyruvate or rapamycin. *Proc Natl Acad Sci.* 2020;117(52):33619-33627. doi:10.1073/pnas.2014213117
- (10) Kim MJ, Lee H, Chanda D, et al. The Role of Pyruvate Metabolism in Mitochondrial Quality Control and Inflammation. *Mol Cells.* 2023;46(5):259-267. doi:10.14348/molcells.2023.2128

- (11) Wang X, Perez E, Liu R, Yan LJ, Mallet RT, Yang SH. Pyruvate Protects Mitochondria from Oxidative Stress in Human Neuroblastoma SK-N-SH Cells. *Brain Res.* 2007;1132(1):1-9. doi:10.1016/j.brainres.2006.11.032
- (12) Tilton WM, Seaman C, Carriero D, Piomelli S. Regulation of glycolysis in the erythrocyte: role of the lactate/pyruvate and NAD/NADH ratios. *J Lab Clin Med.* 1991;118(2):146-152.
- (13) Li X, Yang Y, Zhang B, et al. Lactate metabolism in human health and disease. *Signal Transduct Target Ther.* 2022;7(1):305. doi:10.1038/s41392-022-01151-3
- (14) Zhang ZQ, Xie Z, Chen SY, Zhang X. Mitochondrial dysfunction in glaucomatous degeneration. *Int J Ophthalmol.* 2023;16(5):811-823. doi:10.18240/ijo.2023.05.20
- (15) Ju WK, Perkins GA, Kim KY, Bastola T, Choi WY, Choi SH. Glaucomatous optic neuropathy: Mitochondrial dynamics, dysfunction and protection in retinal ganglion cells. *Prog Retin Eye Res.* 2023;95:101136. doi:10.1016/j.preteyeres.2022.101136
- (16) Jassim AH, Inman DM, Mitchell CH. Crosstalk Between Dysfunctional Mitochondria and Inflammation in Glaucomatous Neurodegeneration. *Front Pharmacol.* 2021;12. doi:10.3389/fphar.2021.699623
- (17) Yang TH, Kang EYC, Lin PH, et al. Mitochondria in Retinal Ganglion Cells: Unraveling the Metabolic Nexus and Oxidative Stress. *Int J Mol Sci.* 2024;25(16):8626. doi:10.3390/ijms25168626

<https://doi.org/10.7554/eLife.105576.2.sa0>